

# **BSc (Hons) Artificial Intelligence and Data Science**

**Module: CM2601 Object Oriented**

**Development**

**Individual Coursework Report**

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# 1.Introduction

## 1.1: Problem Identification

TeamMate: Intelligent-Team-Formation-System solves the common challenges university-gaming-clubs face when forming fair and competitive teams. Traditional-methods often result in uneven skill-levels, personality-conflicts, repeated-roles, clashing-game-preferences, and a heavy-administrative workload-especially with large participant numbers. These issues reduce tournament-quality, lower-participant-satisfaction, and create unnecessary effort for organizers.

## 1.2: Problem Solution

The proposed solution is a fully-automated-system designed to create balanced teams through personality assessment, algorithm-team-formation, and structured-data-handling.

### ☐ Core-components:

- Personality-classification
- Intelligent-Team-Formation
- Dual-Portal-Architecture
- Data-Driven-Workflow

## 1.3: Scope-of-the-Project

### ☐ In-Scope

- Personality-survey-processing and classification
- CSV-based data import/export
- Configurable-team-size
- Smart-team-balancing
- Dual-user-interfaces
- Concurrent-processing-for-efficiency
- Robust-validation and error-handling

### ☐ Out-of-Scope

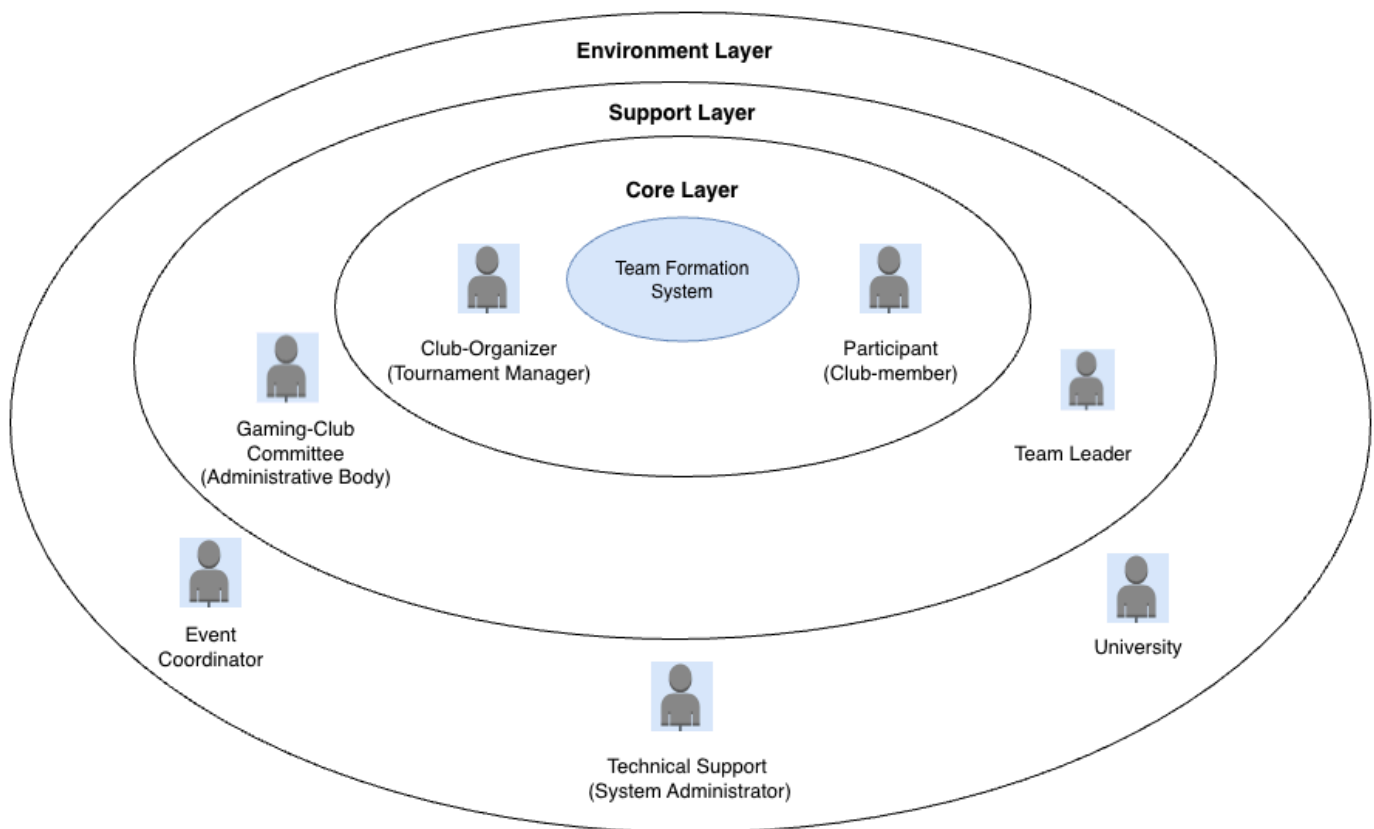
- User-authentication(assumes-Participants-as-already-registered-club-members)
- Tournament-scheduling
- External-database-connectivity

## 1.4:Project Time Lines

- ☐ Weeks 1-2 : UML-design-and-requirement-analysis
- ☐ Weeks 2-5 : Core-OOP-Code-Implementation
- ☐ Weeks 5 : Module-integration-and-initial-testing
- ☐ Weeks 6-7 : Concurrency, error-handling, UI-enhancement
- ☐ Weeks 7-8 : Comprehensive-testing-and-documentation

## 2.Requirement Analysis

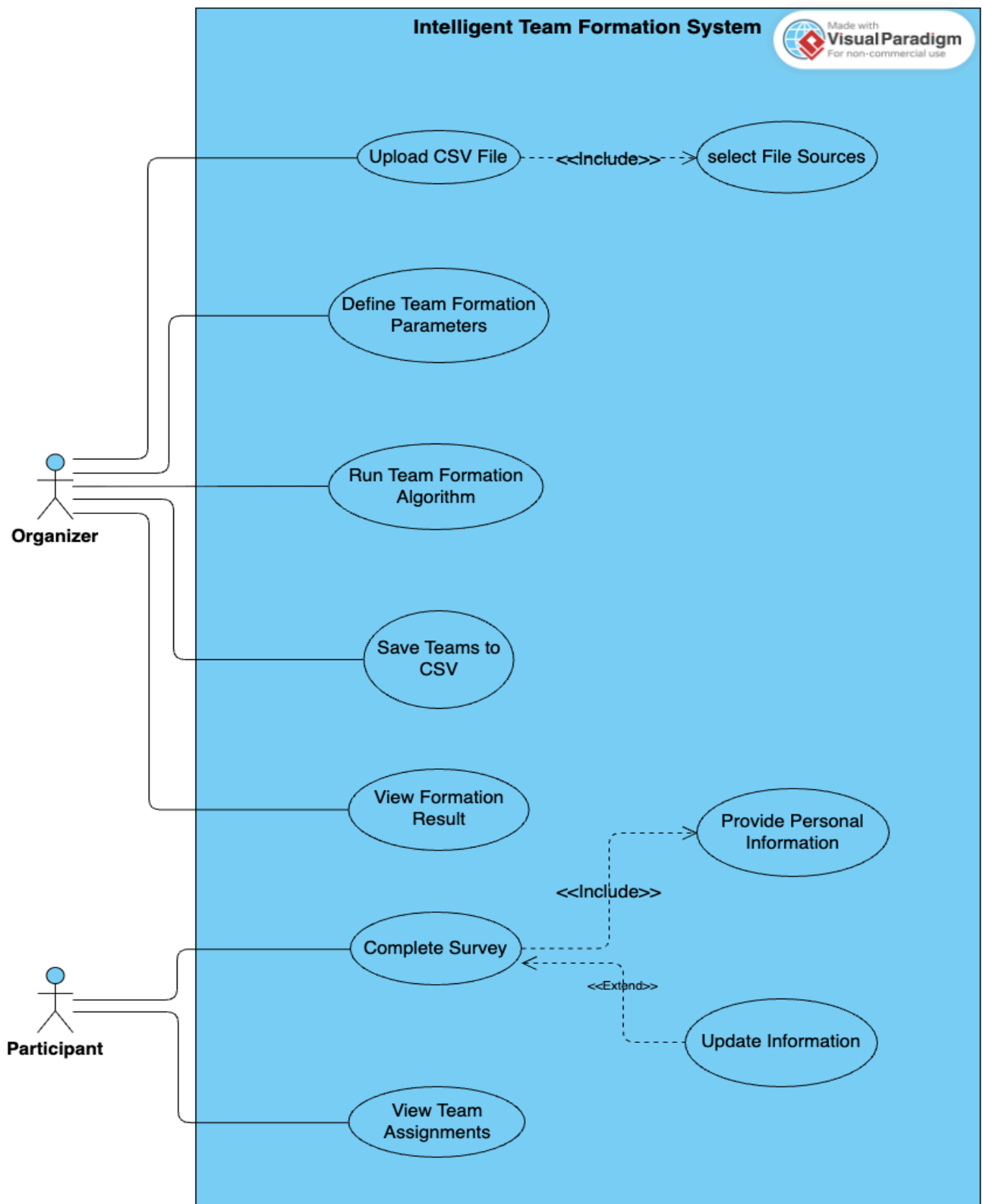
### 2.1 Stake-Holder-Identification



### 2.1.1 Stake-Holder-Requirements

| Club-Organizer                   | Participants          |
|----------------------------------|-----------------------|
| Upload-Participant-Details-CSV   | Complete-Survey       |
| Define-Team-Formation-Parameters | View-Team-Assignments |
| Run-Team-Formation-Algorithm     |                       |
| View-Formation-Result            |                       |
| Save-Formed-Teams-to-CSV         |                       |

## 2.2 Use Case Diagram



## 2.3 Use case Description

### 2.3.1-Upload-CSV-File

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Upload CSV File                                                                                                                                                                                           |
| <b>Section :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Main                                                                                                                                                                                                      |
| <b>Actor :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Organizer                                                                                                                                                                                                 |
| <b>Overview :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | The organizer uploads a CSV file containing participant data to the system for team formation                                                                                                             |
| <b>Purpose :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | To import participant data into the system for processing and team formation                                                                                                                              |
| <b>Preconditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The organizer is authenticated and has a valid CSV file with participant data                                                                                                                             |
| <b>PostConditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Participant data is successfully loaded and validated in the system                                                                                                                                       |
| <b>Main Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                           |
| <ol style="list-style-type: none"><li>1. Organizer access to the organizer portal through main menu</li><li>2. The organizer selects the Upload CSV file option</li><li>3. System Prompt for file source selection</li><li>4. The organizer selects the CSV file or provide file path from local storage</li><li>5. The system validates the file format and data</li><li>6. The system confirms successful upload with log message and displays upload summary with participant count and validation results</li><li>7. System stores participant data in memory for team formation</li></ol> |                                                                                                                                                                                                           |
| <b>Alternative Flow :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>If the file format is invalid, the system displays an error message and prompts for a new file</p> <p>If required data fields are missing, the system highlights the error and request corrections</p> |
| <b>Includes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Select file Sources                                                                                                                                                                                       |
| <b>Extends :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                                                                                                                                                                                                      |

### **2.3.2-Define-Team-Formation-Parameters**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Define team formation parameters                                                                                                                                                                                                                |
| <b>Section :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Main                                                                                                                                                                                                                                            |
| <b>Actor :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Organizer                                                                                                                                                                                                                                       |
| <b>Overview :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The organizer sets team size and formation constraints before running the algorithm                                                                                                                                                             |
| <b>Purpose :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | To configure how teams should be formed based on tournament requirements                                                                                                                                                                        |
| <b>Preconditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Participant data CSV file successfully uploaded to the system and validated                                                                                                                                                                     |
| <b>PostConditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Team formation parameters are set and ready for algorithm execution                                                                                                                                                                             |
| <b>Main Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                 |
| <ol style="list-style-type: none"><li>1. The organizer access Set Team Formation Parameters through organizer action menu</li><li>2. The system displays configurable options (team size, constraints)</li><li>3. The organizer sets team size (N)</li><li>4. System validates team size against participant count</li><li>5. The organizer optionally adjusts constraints weights</li><li>6. The system saves the parameters and display configuration summary</li></ol> |                                                                                                                                                                                                                                                 |
| <b>Alternative Flow :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>If invalid team size is entered( too large/ too small) the system will display the ranges we can formed teams</p> <p>If constraints weight exceed the limit system will automatically adjust it and display it for the organizer to view</p> |
| <b>Includes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | None                                                                                                                                                                                                                                            |
| <b>Extends :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | None                                                                                                                                                                                                                                            |

### **2.3.3-Run-Team-Formation-Algorithm**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Run team formation algorithm                                                                                                                                                                                 |
| <b>Section :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Main                                                                                                                                                                                                         |
| <b>Actor :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Organizer                                                                                                                                                                                                    |
| <b>Overview :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | The system process participant data to form balanced teams based on personality type, skills, and preferences                                                                                                |
| <b>Purpose :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | To automatically create well-balanced teams that meet diversity criteria for optimal team performance                                                                                                        |
| <b>Preconditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Team size parameters are defined and validated.                                                                                                                                                              |
| <b>PostConditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Teams are formed and ready for viewing or saving                                                                                                                                                             |
| <b>Main Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                              |
| <ol style="list-style-type: none"><li>1. The organizer initiates team formation</li><li>2. System retrieve team formation parameters and participant data</li><li>3. The system classifies all participants by personality type using personality classifier</li><li>4. The system applies advanced matching algorithm<br/>using-parallel-processing-for-constraint-satisfaction:<ol style="list-style-type: none"><li>a.Distributes leaders first (max 1 per team)</li><li>b.Consider game variety (max 2 same games per team)</li><li>c.Ensures role diversity (min 3 different roles)</li><li>d.Balances skill level across teams</li><li>e.Optimizes personality mis(Leaders + Thinkers + Balanced)</li></ol></li><li>5. The system forms teams using constraints satisfaction approach</li><li>6. System validate team composition against all constraints</li><li>7. System generates team statistics and balanced report</li><li>8. System will store formed teams in new csv file</li><li>9. The system displays the formed teams and statics to the organizer</li></ol> |                                                                                                                                                                                                              |
| <b>Alternative Flow :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | If algorithm cannot meet all constraints, it forms the best possible teams and reports compromises, provides detailed constraints violation report, and organizer can choose to adjust parameters and re-run |
| <b>Includes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | None                                                                                                                                                                                                         |
| <b>Extends :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | None                                                                                                                                                                                                         |

### 2.3.4-Save-Teams-to-CSV-File

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Save Teams to CSV                                                                                                                                                                                                                                               |
| <b>Section :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Main                                                                                                                                                                                                                                                            |
| <b>Actor :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Organizer                                                                                                                                                                                                                                                       |
| <b>Overview :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | The organizer saves the formed teams to a CSV file for external use or record keeping                                                                                                                                                                           |
| <b>Purpose :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | To export the team formations results in a portable format for tournament organization or sharing                                                                                                                                                               |
| <b>Preconditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Teams have been successfully formed and are displayed to the organizer                                                                                                                                                                                          |
| <b>PostConditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Teams are saved in a CSV file at the specified location                                                                                                                                                                                                         |
| <b>Main Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                 |
| <ol style="list-style-type: none"><li>1. The organizer selects the save team option from menu option</li><li>2. System checks team formation status and validates teams exist</li><li>3. The system prompts for file location and default file name</li><li>4. The organizer specifies the save location and file name</li><li>5. System validates write permission and path availability</li><li>6. The system generates CSV with team assignments</li><li>7. System writes formed CSV data to specified location</li><li>8. The system confirms successfully save with summary statistics</li></ol> |                                                                                                                                                                                                                                                                 |
| <b>Alternative Flow :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>If the save location is invalid, the system displays specific error message and prompts for a new location</p> <p>If file writing fails, the system displays detailed error message with cause, allows retry operation with same or different parameters</p> |
| <b>Includes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None                                                                                                                                                                                                                                                            |
| <b>Extends :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | None                                                                                                                                                                                                                                                            |

### **2.3.5-View-Formation-Result**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | View formation result                                                                                      |
| <b>Section :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Main                                                                                                       |
| <b>Actor :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Organizer                                                                                                  |
| <b>Overview :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | The organizer views the teams formed by the algorithm with detailed participant information                |
| <b>Purpose :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | To view and analyze the team composition before finalizing or saving                                       |
| <b>Preconditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Team formation algorithm has been successfully executed and Teams have been formed                         |
| <b>PostConditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | The organizer has reviewed the team assignments                                                            |
| <b>Main Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                            |
| <ol style="list-style-type: none"><li>1. The organizer selects to view formation results</li><li>2. System checks team formation status</li><li>3. System retrieves latest formed data from memory</li><li>4. System load team compositions and member details</li><li>5. The system generate team analysis report</li><li>6. The system displays all formed details with member details</li><li>7. The system shows team statistics( average skill, personality mix, role distribution)</li><li>8. The organizer reviews the teams and analyze compositions</li><li>9. The organizer can navigate between teams and view details</li><li>10. Organizer evaluate team composition and make decisions</li></ol> |                                                                                                            |
| <b>Alternative Flow :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | If teams are not optimally balanced, the organizer can choose to re-run formation with adjusted parameters |
| <b>Includes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | None                                                                                                       |
| <b>Extends :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | None                                                                                                       |

### **2.3.6-Complete-Survey**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Complete Survey                                                                                                                                                                                                                                                                                                                                     |
| <b>Section :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Main                                                                                                                                                                                                                                                                                                                                                |
| <b>Actor :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Participant                                                                                                                                                                                                                                                                                                                                         |
| <b>Overview :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Participants complete a personality and preference survey to provide data for team formation                                                                                                                                                                                                                                                        |
| <b>Purpose :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | To gather essential participant data including personality traits, game preferences, and role choices                                                                                                                                                                                                                                               |
| <b>Preconditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | The Participant registered as a club member<br>The participants has access to the survey system                                                                                                                                                                                                                                                     |
| <b>PostConditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The participant's data is recorded in the system for team formation                                                                                                                                                                                                                                                                                 |
| <b>Main Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                     |
| <ol style="list-style-type: none"><li>1. Participant access the Participant portal through the Main menu</li><li>2. The participant select the complete survey option from member portal</li><li>3. The system verifies member status(through id) and load survey template</li><li>4. System prompt for personal information (id,name, email,phonenumber)</li><li>5. Participant Provide required personal informations</li><li>6. System validate the personal informations</li><li>7. The system presents personality questions (Q1-Q5) with 1-5 rating scale</li><li>8. The participant rate each statement from 1-5</li><li>9. System Validate each rating</li><li>10. The participant select preferred game from available options</li><li>11. System validate each input weather participant choose from the given selections</li><li>12. Participant chooses preferred role with role descriptions displayed</li><li>13. System validate each input weather participant choose from the given selections</li><li>14. The participant provides skill level information using slider(1-10)</li><li>15. System validates skill level</li><li>16. The system calculates personality score and type</li><li>17. System Display Details newly added</li><li>18. The system saves the complete survey data</li><li>19. System Displays completion confirmation</li></ol> |                                                                                                                                                                                                                                                                                                                                                     |
| <b>Alternative Flow :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>If the participant provide invalid information(Personal Information,ratings, skill level,Games,Roles) ,system will display error message and prompt for correction</p> <p>If participant enter a already existed id then system will prompt weather need to update system or not if yes then new information will be updated and save to csv</p> |

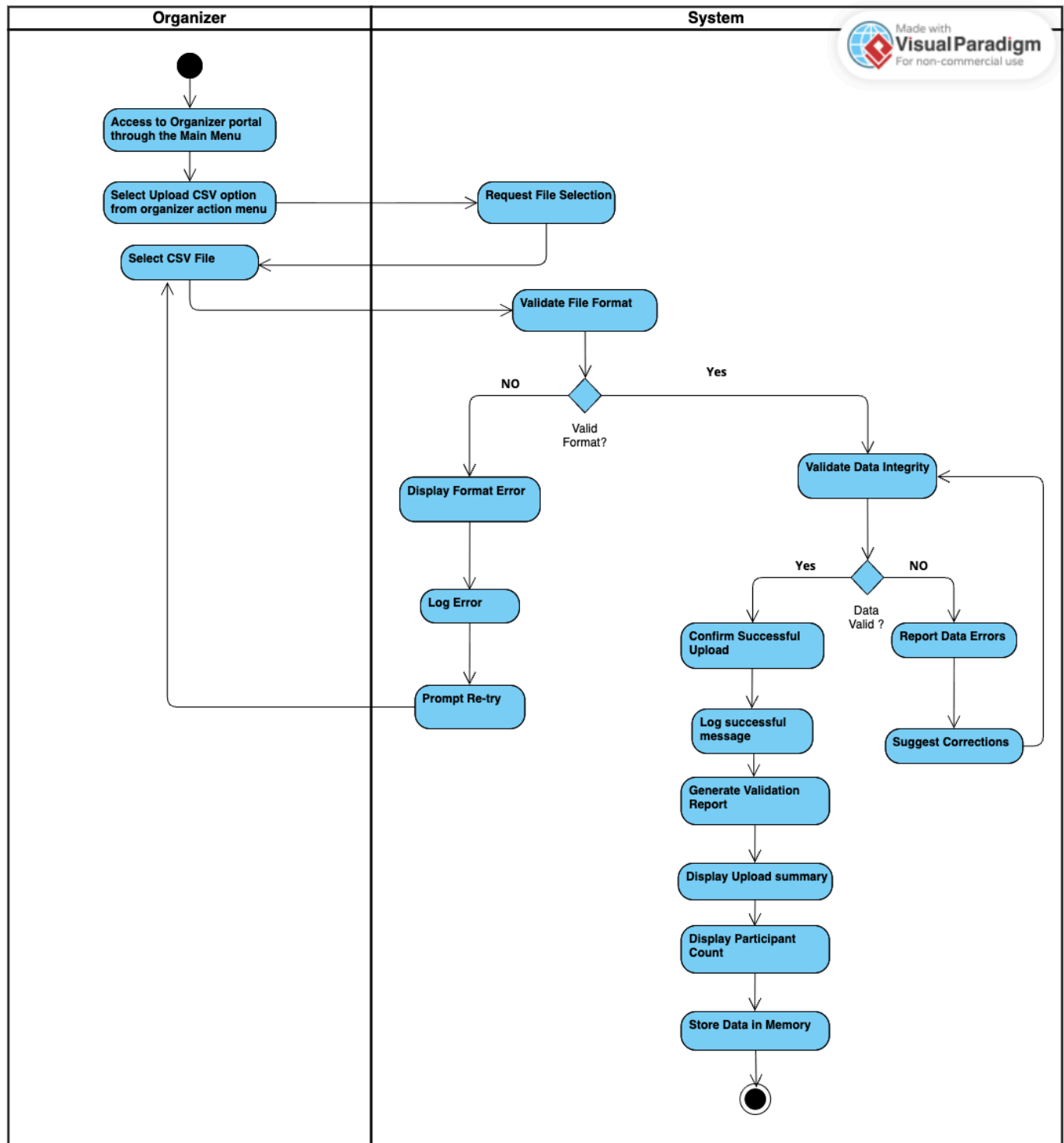
|                        |                                                                                                                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b> | Complete Survey                                                                                                                                                                                                                                        |
| <b>Section :</b>       | Main                                                                                                                                                                                                                                                   |
|                        | <p>If the participant provides incomplete responses, the system will continue to loop that question till the answer is complete and correct</p> <p>If technical issues occur during submission, the system allows retry and preserves entered data</p> |
| <b>Includes :</b>      | Provide Personal Informations                                                                                                                                                                                                                          |
| <b>Extends :</b>       | Update Information                                                                                                                                                                                                                                     |

### **2.3.7-View-Team-Assignment**

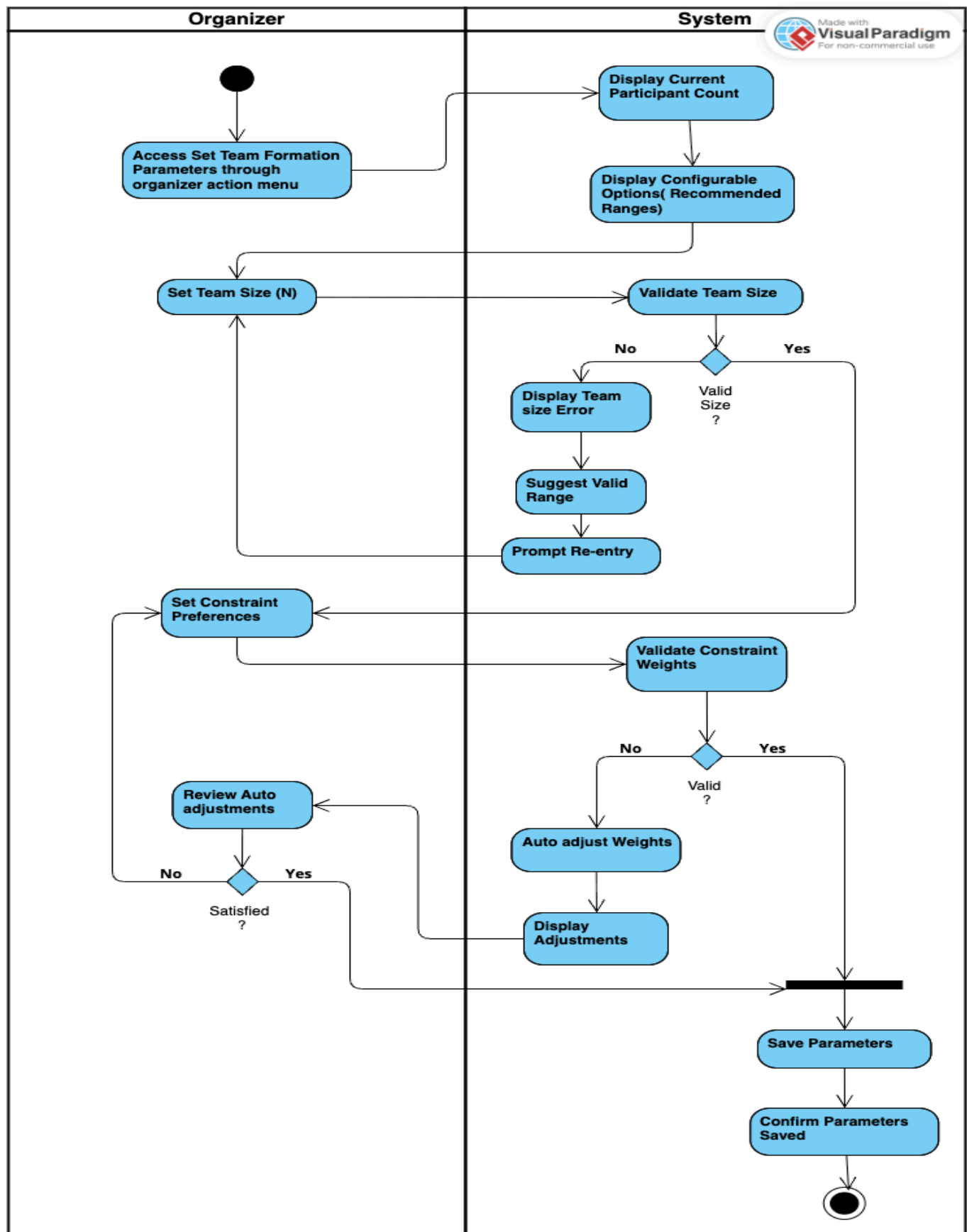
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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case Name :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | View Team Assignments                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Section :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Main                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Actor :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Participant                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Overview :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Participant views in which team they have been assigned by system for upcoming events                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | To allow participants to see their team assignments and basic team mate informations for coordination                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Preconditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Participant has completed the survey<br>Teams have been formed by the system                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>PostConditions :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Participants knows their team assignments and can contact their team mates                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Main Flow</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <ol style="list-style-type: none"><li>1. Participant access the Participant portal through Main menu</li><li>2. Participant select view team assignment option</li><li>3. System check team formation status and validate team exists</li><li>4. System prompt for participant identification (ID)</li><li>5. System authenticate participant identity through search and validation</li><li>6. System retrieves the participant's specific team assignment from mapping</li><li>7. System loads comprehensive team details</li><li>8. System displays formatted team assignment with all relevant details</li><li>9. Participant reviews their team composition and teammate information</li></ol> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Alternative Flow :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>If no teams haven't been formed yet, system displays "No Team have been formed yet" message and suggest contacting organizer to run team formation</p> <p>If participant has no assignment, system displays "No team assignments found" message with search suggestions and shows available participant IDs for reference</p> <p>If entered participant Id doesn't exist or have no valid format ,system will suggest what to do next, and what format we need to add Ids</p> |
| <b>Includes :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Extends :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 3. Activity Diagrams

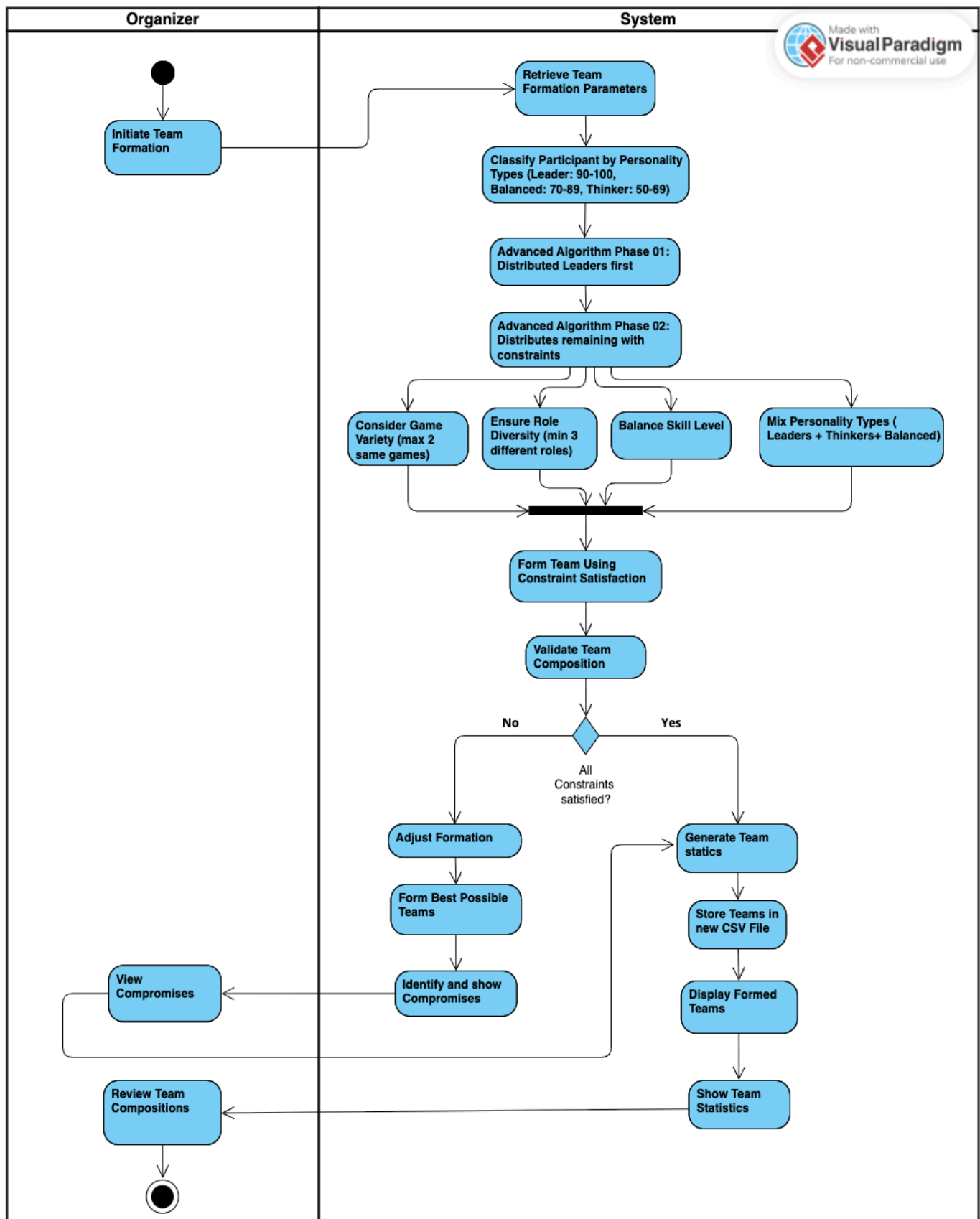
### 3.1-Upload-CSV-File



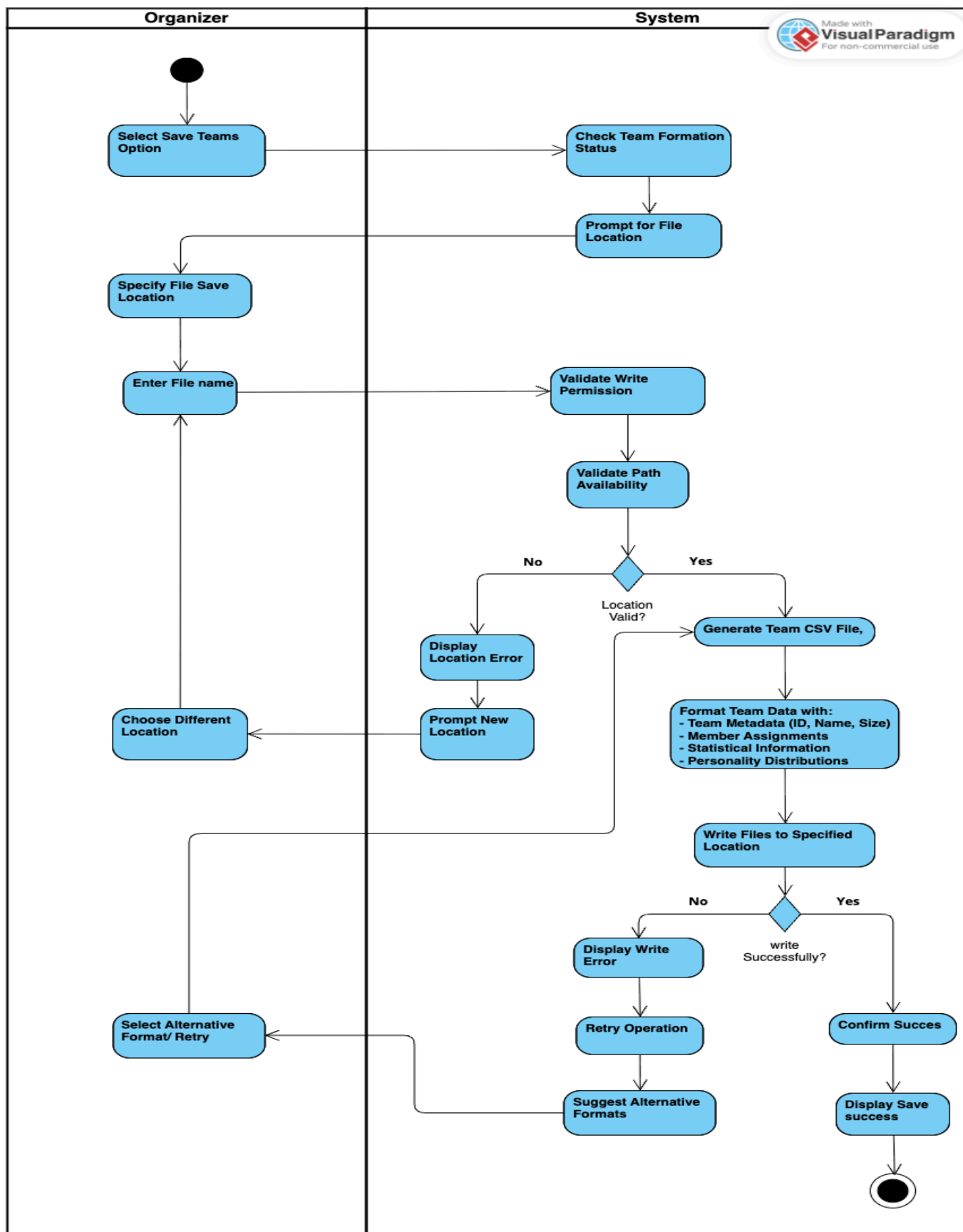
### 3.2-Define-Team-Formation-Parameters



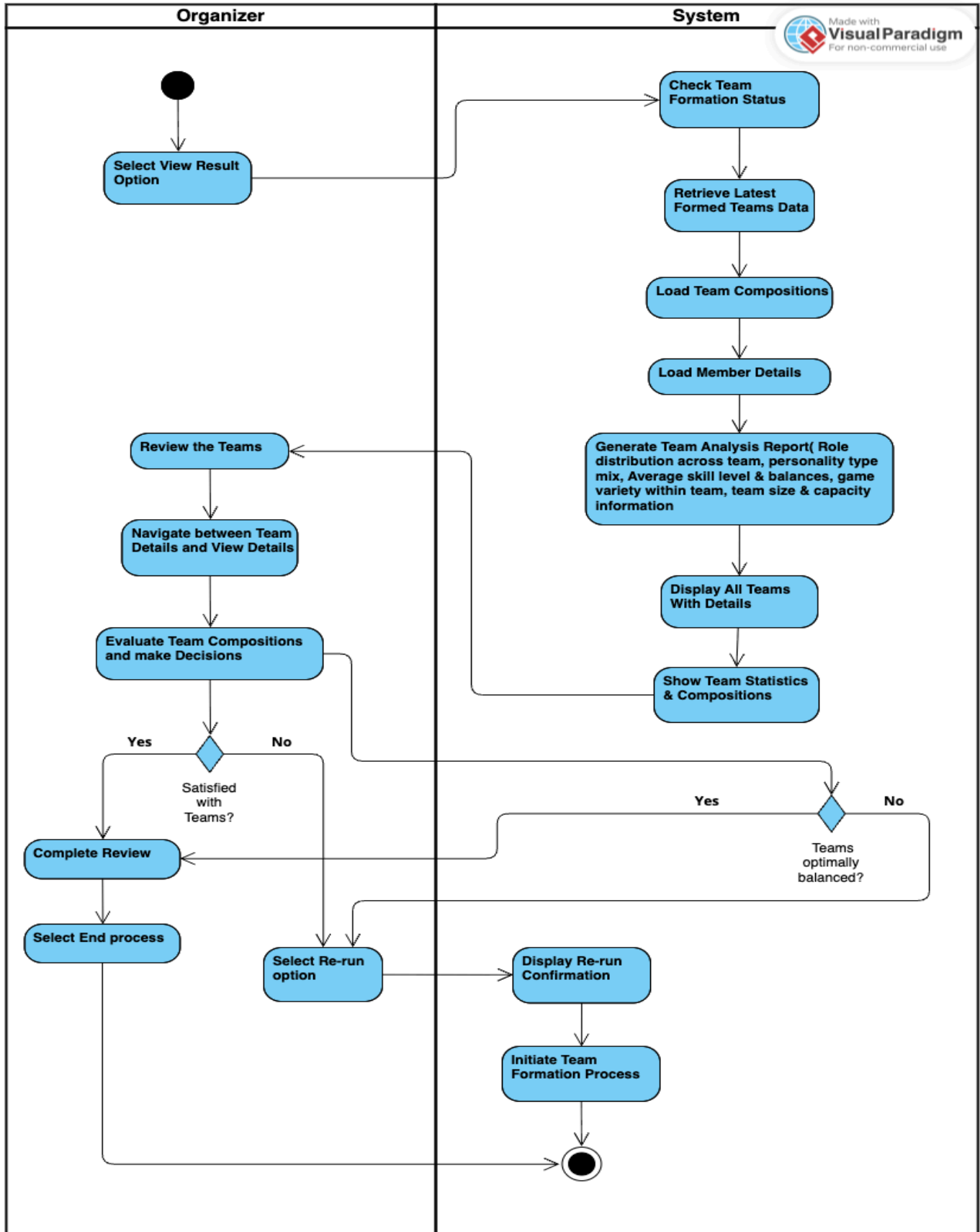
### 3.3-Run-Team-Formation-Algorithm



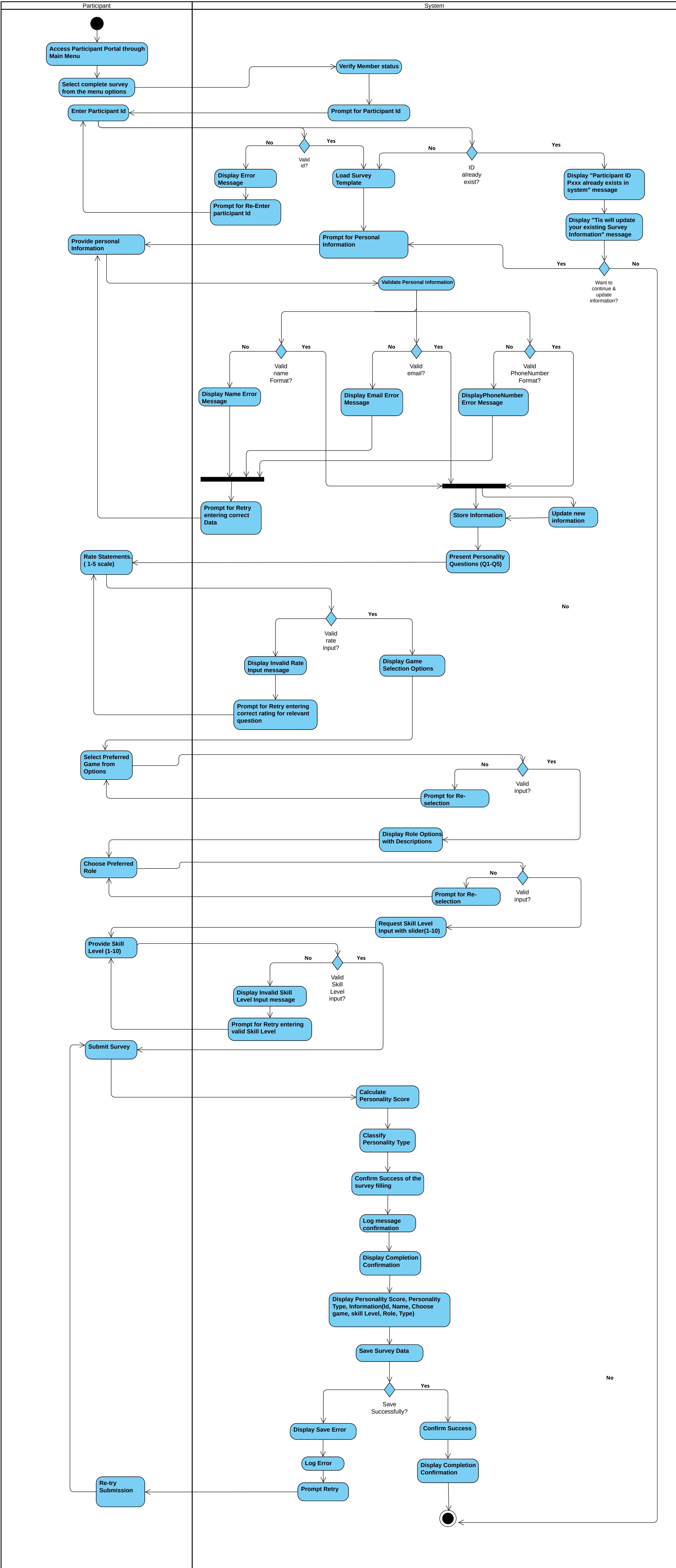
### 3.4-Save-Teams-to-CSV-File



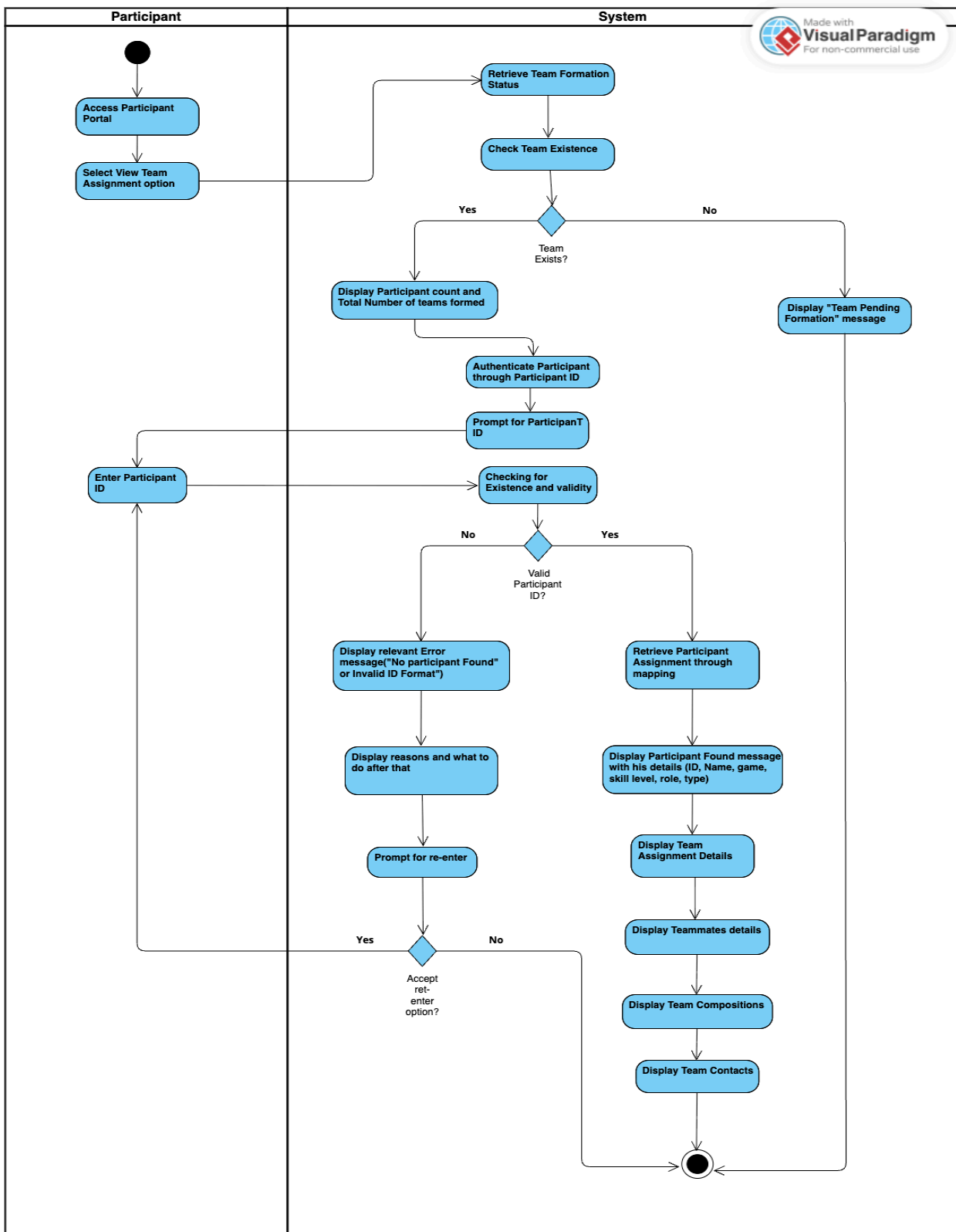
### 3.5-View-Formation-Result



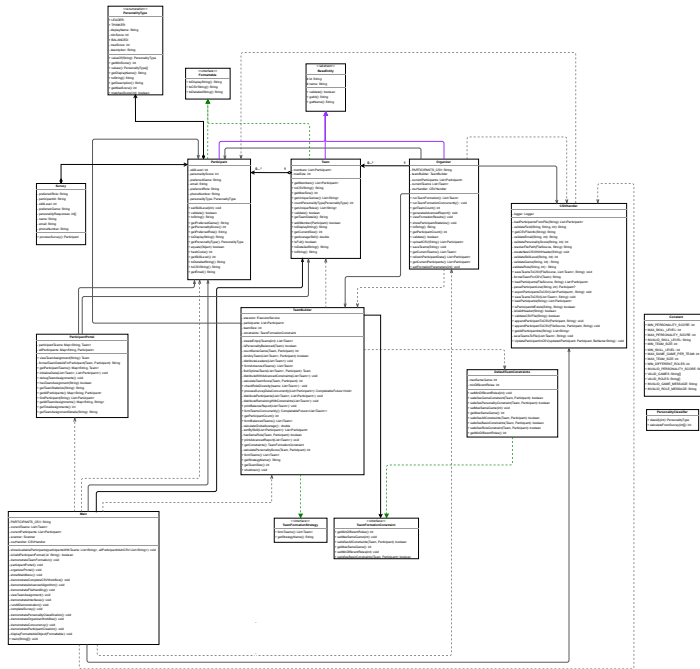
3.6:Complete Survey.



### 3.7-View Team Assignments

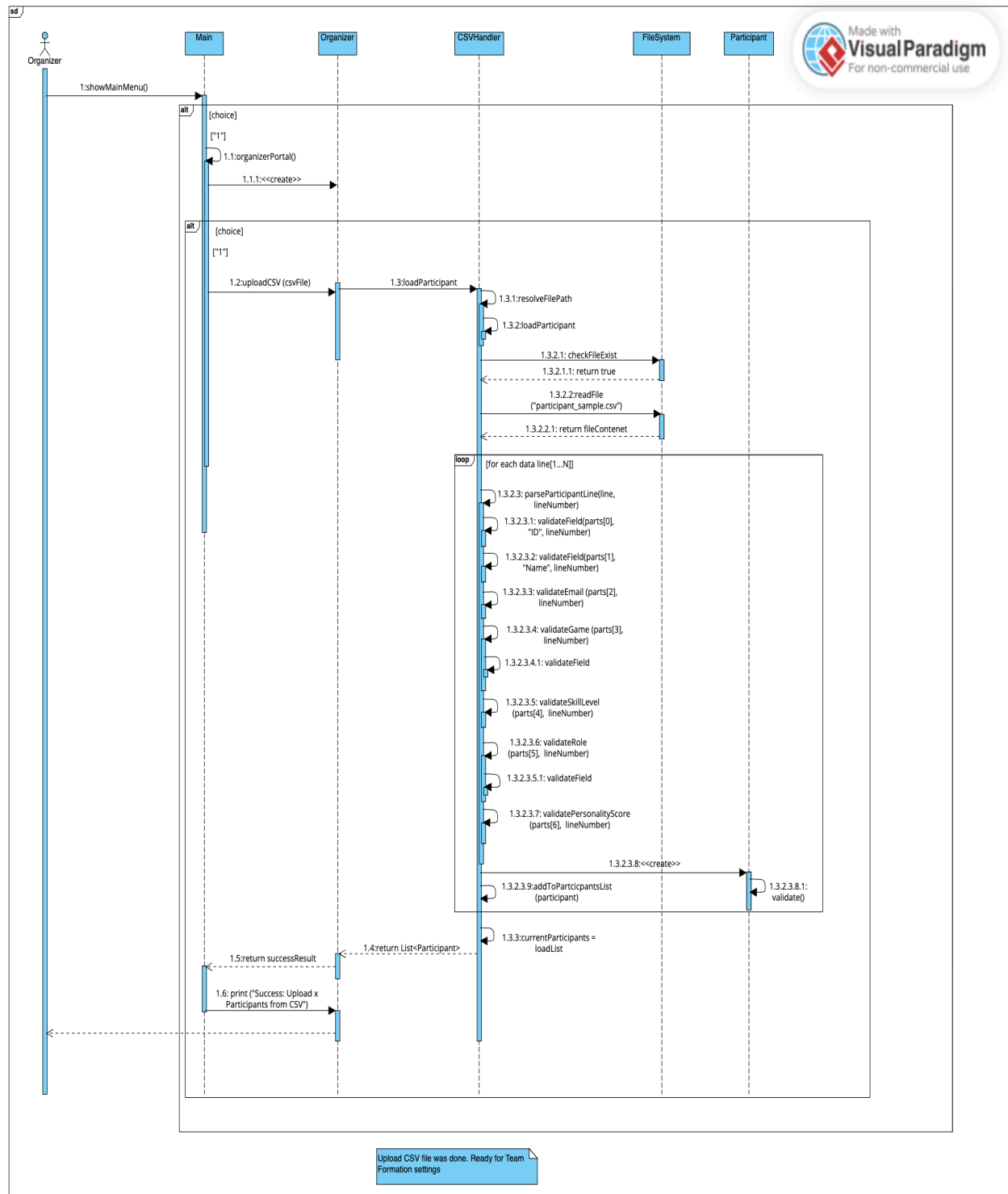


#### 4. Class Diagram

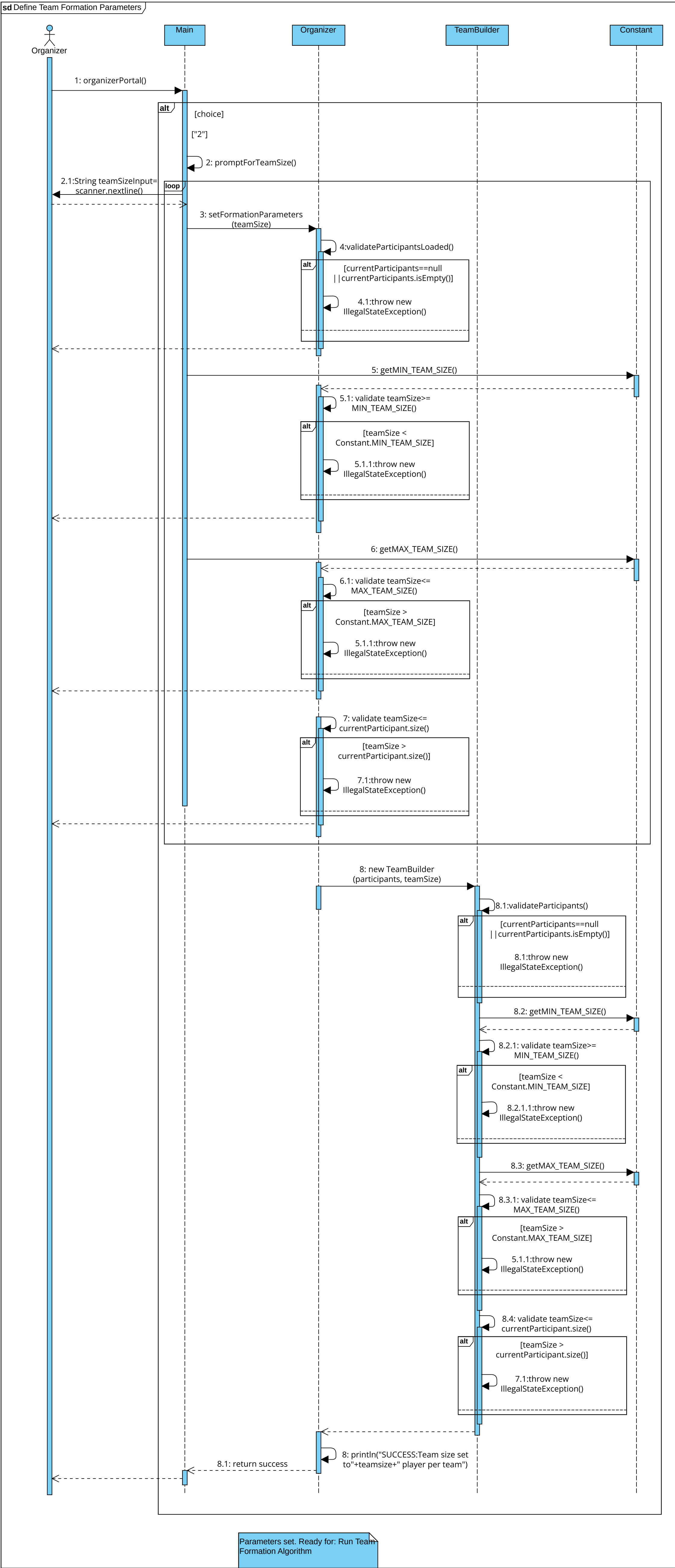


## 5. Sequence Diagrams

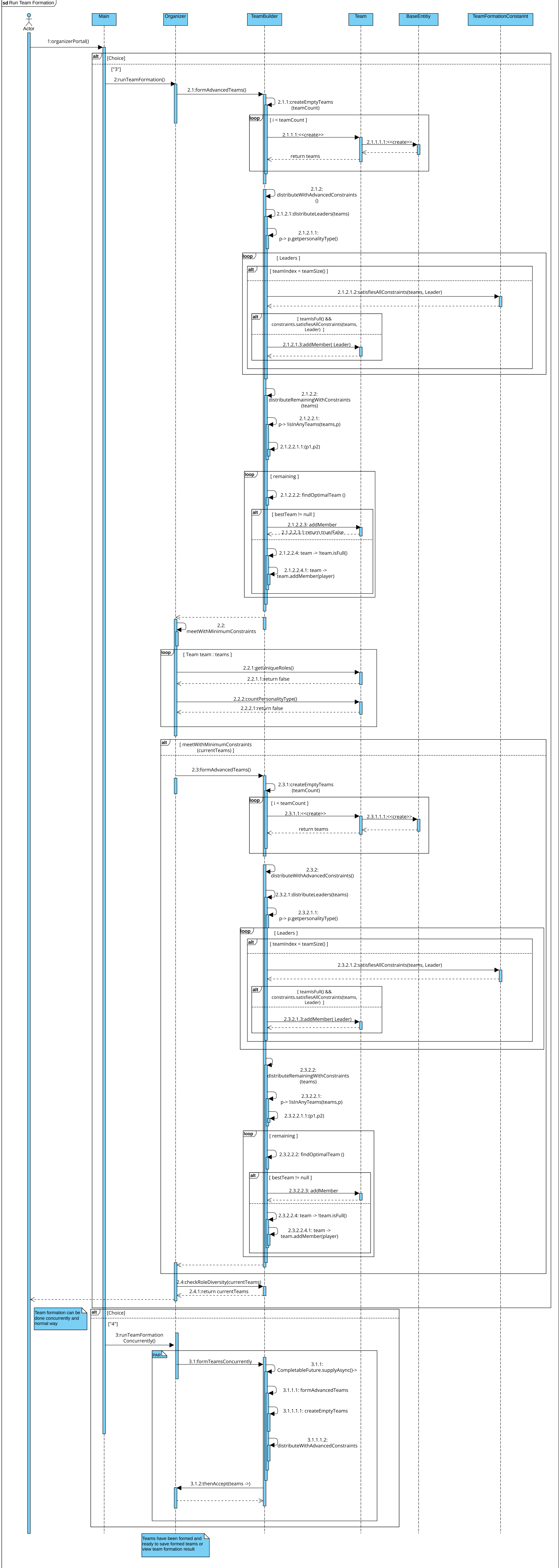
### 5.1-Upload-CSV-File



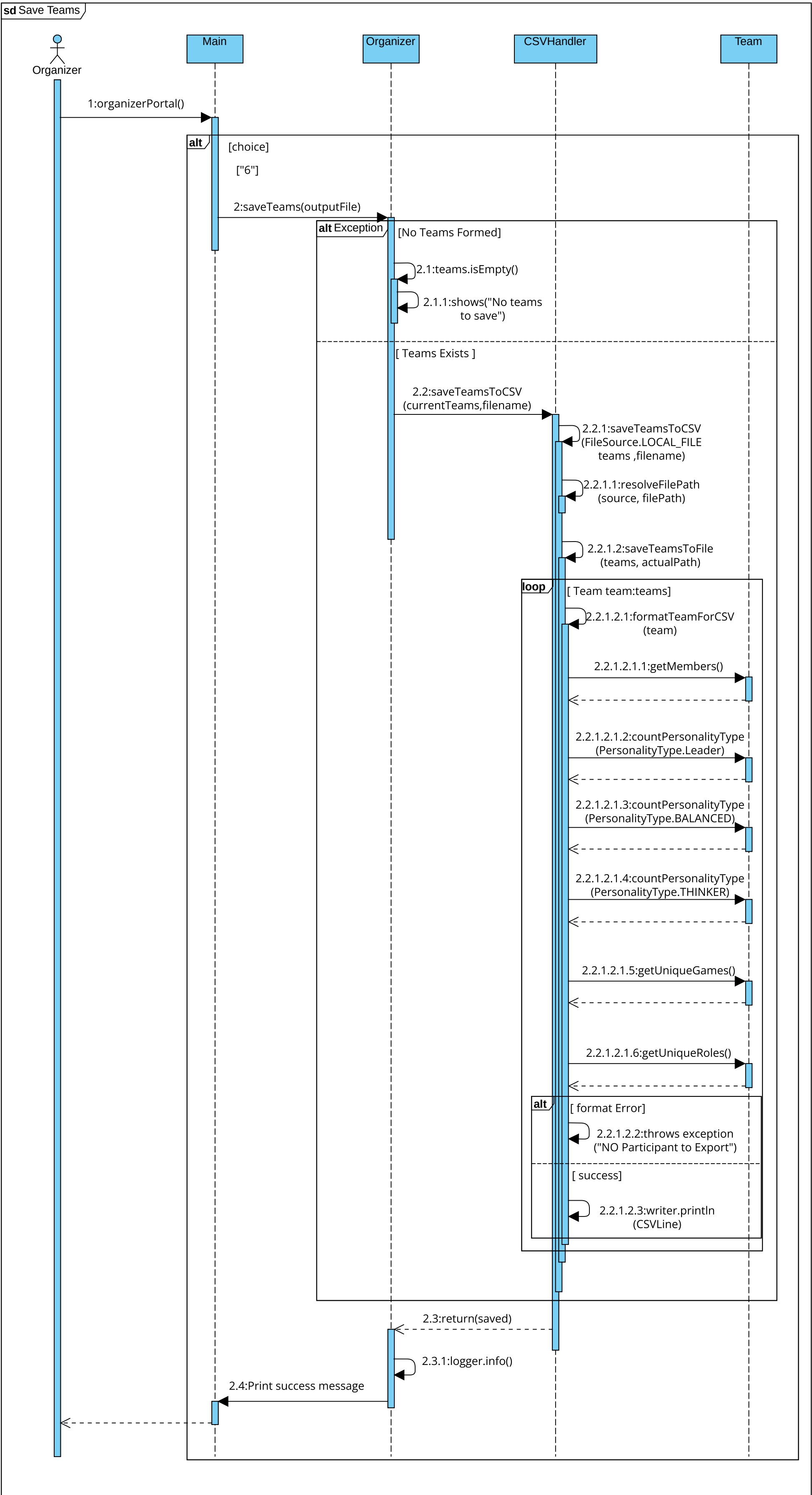
## **5.2-Define-Team-Formation-Parameters**



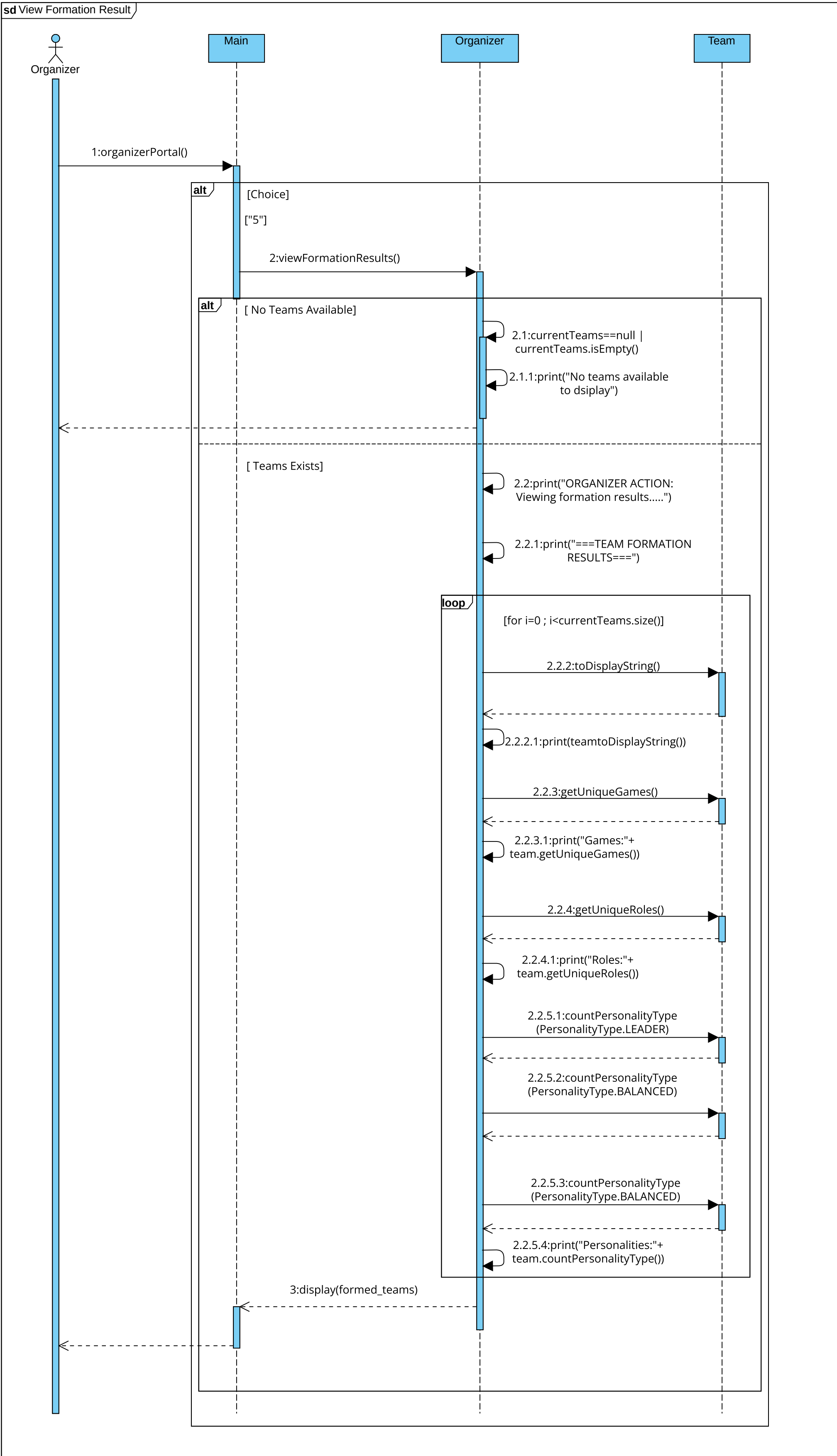
### **5.3-Run-Team-Formation-Algorithm**



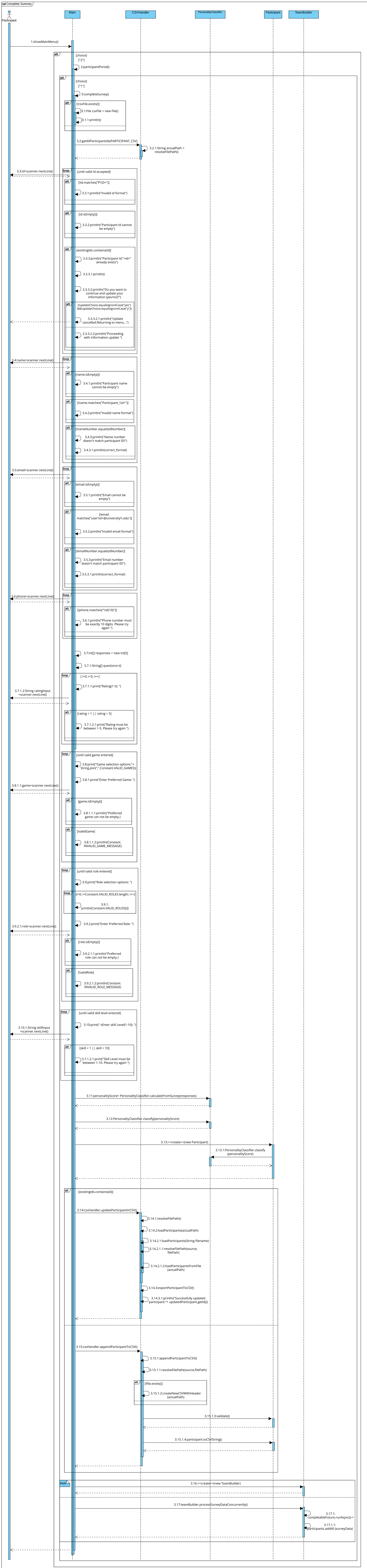
## **5.4-Save-Teams-to-CSV-File**



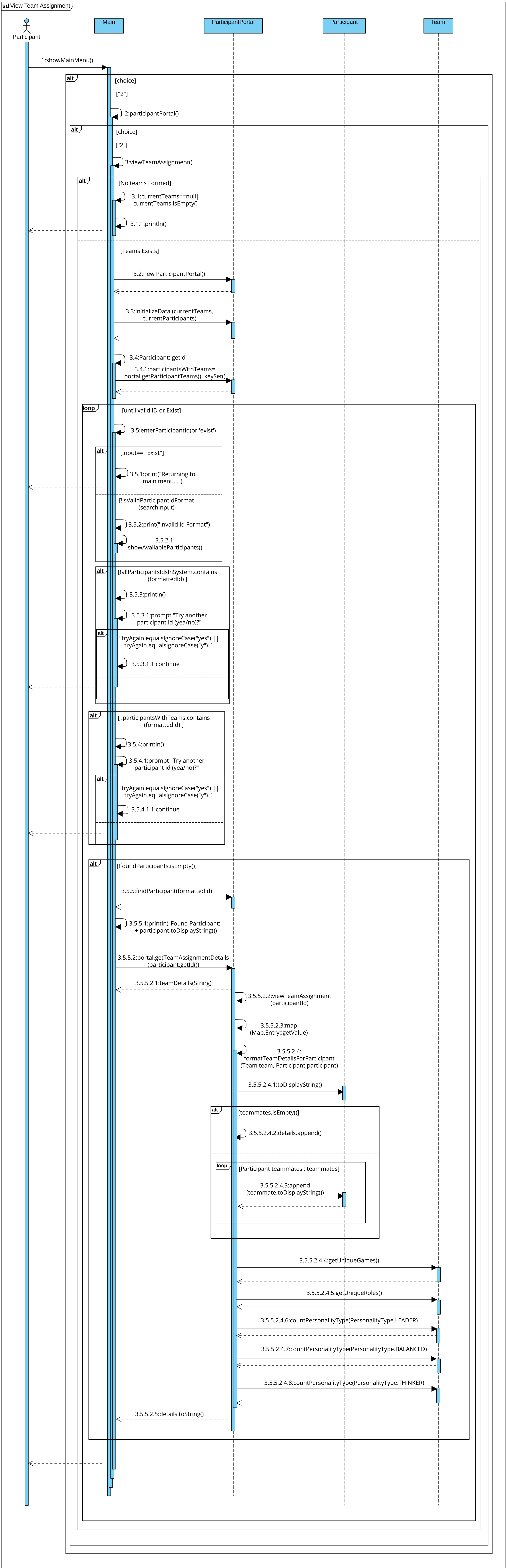
## **5.5-View-Formation-Result**



## **5.6 Complete-Survey**



## **5.7-View-Team-Assignment**



## 6. Testing and Evaluation

Testing was conducted through console-based-functional-demonstration and structured JUnit-tests. Manual-tests validated classification, CSV-Handling,team-formation, and concurrency-behavior, while automated-units and integration-tests-verified constraints, error-handling, workflow-correctness, and-overall-system reliability.

### Manual-Testing

#### Personality-classification-Demonstration

```
private static void demonstratePersonalityClassification() { 1 usage 2 Yenu0808 *
    System.out.println("\n1. PERSONALITY CLASSIFICATION DEMONSTRATION");
    System.out.println("-----");

    int[] testScores = {95,80,65,45,100,75};

    for (int score: testScores) {
        try{
            var personality = PersonalityClassifier.classify(score);
            System.out.printf("Score %3d -> %-8s(Range: %d-%d)%n",score,personality,personality.getMinScore()
                               ,personality.getMaxScore());
        }catch (IllegalArgumentException e){
            System.out.printf("Score %3d -> ERROR: %s%n", score, e.getMessage());
        }
    }
    System.out.println();
}
```

=== SYSTEM DEMONSTRATIONS ===

#### 1. PERSONALITY CLASSIFICATION DEMONSTRATION

```
-----
Score  95 -> Leader   (Range: 90-100)
Score  80 -> Balanced(Range: 70-89)
Score  65 -> Thinker (Range: 50-69)
Score  45 -> ERROR: Invalid score: 45.Must be 0-100
Score 100 -> Leader   (Range: 90-100)
Score  75 -> Balanced(Range: 70-89)
```

## Participant-Creation-Demo

```
private static void demonstrateParticipantCreation() { 1 usage 8 Yenuli0808 *
    System.out.println("\n2. PARTICIPANT CREATION DEMONSTRATION");
    System.out.println("-----");

    try{
        Participant participant1 = new Participant( id: "P001", name: "John Doe", email: "john@university.edu", phoneNumber: "0731093108",
            preferredGame: "Valorant", skillLevel: 8, preferredRole: "Strategist", personalityScore: 85);
        System.out.println("Participant 1 created: " + participant1);

        try{
            Participant invalid= new Participant( id: "P002", name: "Jane Smith", email: "jane@university.edu", phoneNumber: "0701053109",
                preferredGame: "CS:GO", skillLevel: 15, preferredRole: "Attacker", personalityScore: 85);
        }catch (IllegalArgumentException e){
            System.out.println("Rejected Invalid Participant: " + e.getMessage());
        }
    }catch (Exception e){
        System.out.println("Error creating Participant: " + e.getMessage());
    }

    System.out.println();
}
```

### 2. PARTICIPANT CREATION DEMONSTRATION

-----

Participant 1 created: P001 - John Doe (Valorant) | Skill: 8 | Role: Strategist | Type: Balanced

Rejected Invalid Participant: Skill Level must be 1-10

## Advanced-Algorithm-Demo

```
private static void demonstrateAdvancedAlgorithm(){ 1 usage 2 Yenu0808 *

    System.out.println("\n6. ADVANCED ALGORITHM DEMONSTRATION");
    System.out.println("-----");

    List<Participant> participants = Arrays.asList(
        new Participant( id: "P001", name: "Alice", email: "alice@edu.com", phoneNumber: "0711153109", preferredGame: "Valorant", skillLevel: 8,
            preferredRole: "Strategist", personalityScore: 95),
        new Participant( id: "P002", name: "Bob", email: "bob@edu.com", phoneNumber: "0721253109", preferredGame: "Valorant", skillLevel: 6,
            preferredRole: "Attacker", personalityScore: 75),
        new Participant( id: "P003", name: "Charlie", email: "charlie@edu.com", phoneNumber: "0731353109", preferredGame: "Valorant", skillLevel: 7,
            preferredRole: "Defender", personalityScore: 65),
        new Participant( id: "P004", name: "Diana", email: "diana@edu.com", phoneNumber: "0741453109", preferredGame: "CS:GO", skillLevel: 9,
            preferredRole: "Supporter", personalityScore: 85),
        new Participant( id: "P005", name: "Eve", email: "eve@edu.com", phoneNumber: "0751553109", preferredGame: "CS:GO", skillLevel: 5,
            preferredRole: "Coordinator", personalityScore: 92),
        new Participant( id: "P006", name: "Frank", email: "frank@edu.com", phoneNumber: "0761653109", preferredGame: "DOTA 2", skillLevel: 8,
            preferredRole: "Strategist", personalityScore: 88)
    );

    System.out.println("Using Advanced Algorithm with 3 Simple Rules:");
    System.out.println("1. One leader per team maximum");
    System.out.println("2. Maximum 2 players from same game");
    System.out.println("3. Prefer different roles for diversity");

    TeamBuilder advancedBuilder = new TeamBuilder(participants, teamSize: 3);
    List<Team> advancedTeams = advancedBuilder.formAdvancedTeams();

    System.out.println("\nADVANCED ALGORITHM: Formed " + advancedTeams.size() + " optimally balanced teams");
    advancedBuilder.printAdvancedReport(advancedTeams);

    System.out.println("\n--- COMPARISON: Basic vs Advanced ---");
    List<Team> basicTeams = advancedBuilder.formBalancedTeams();
    System.out.println("Basic Algorithm: Focuses only on skill balance");
    System.out.println("Advanced Algorithm: Adds game variety + role diversity + personality mix");
}
```

## 6. ADVANCED ALGORITHM DEMONSTRATION

-----

Using Advanced Algorithm with 3 Simple Rules:

1. One leader per team maximum
2. Maximum 2 players from same game
3. Prefer different roles for diversity

Applying Advanced Constraints:

1. One leader per team maximum
2. Max 2 players from same game
3. At least 3 different roles
4. Balanced personality mix

Existing leaders: 0, Adding leader: true

Existing leaders: 0, Adding leader: true

ADVANCED ALGORITHM: Formed 2 optimally balanced teams

=== ADVANCED TEAM ANALYSIS ===

Constraints Applied: Leader limit, Game variety, Role diversity, Personality mix

Team 1:

Players: 3/3

Avg Skill: 7.0

Personalities: 1L 1B 1T

Games: [Valorant]

Roles: [Strategist, Defender, Attacker]

Constraints: ✓Leader ✗Games ✓Roles ✓Personality

Team 2:

Players: 3/3

Avg Skill: 7.3

Personalities: 1L 2B 0T

Games: [CS:GO, DOTA 2]

Roles: [Coordinator, Supporter, Strategist]

Constraints: ✓Leader ✓Games ✓Roles ✓Personality

## CSV-File-Handling-Demo

```
private static void demonstrateFileHandling() { 1 usage 2 Yenuli0808
    System.out.println("\n5. FILE HANDLING DEMONSTRATION");
    System.out.println("-----");

    try{
        CSVHandler csvHandler = new CSVHandler();

        System.out.println("Loading Participant from CSV file...");
        List<Participant> participants = csvHandler.loadParticipants( filename: "participants_sample.csv");
        System.out.printf("Successfully loaded %d participants from CSV file%n", participants.size());

        TeamBuilder builder = new TeamBuilder(participants, teamSize: 4);
        List<Team> teams = builder.formBalancedTeams();

        System.out.println("Saving teams to output CSV...");
        csvHandler.saveTeamsToCSV(teams, filename: "formed_teams-.csv");
        System.out.println("SUCCESS: Saved " + teams.size() + " teams to 'formed_teams-.csv'");

        System.out.println("\nSample of loaded data (first 3 participants):");
        for (int i = 0; i < Math.min(3, participants.size()); i++) {
            System.out.println("  - " + participants.get(i).toString());
        }
    }catch (Exception e){
        System.out.println("FILE HANDLING ERROR: " + e.getMessage());
        System.out.println("Note: Make sure 'participants_sample.csv' is in your project root directory");
    }
}
```

### 5. FILE HANDLING DEMONSTRATION

```
-----
Loading Participant from CSV file...
Dec 04, 2025 12:48:32 PM Service.CSVHandler loadParticipants
INFO: Loading participants from: participants_sample.csv (Source: LOCAL_FILE)
Dec 04, 2025 12:48:32 PM Service.CSVHandler loadParticipantsFromFile
INFO: Successfully loaded 100 participants from participants_sample.csv
Dec 04, 2025 12:48:32 PM Service.CSVHandler saveTeamsToCSV
INFO: Saving 25 teams to: formed_teams-.csv
Dec 04, 2025 12:48:32 PM Service.CSVHandler saveTeamsToFile
INFO: Successfully saved 25 teams to formed_teams-.csv
Loading from: participants_sample.csv
Warning: CSV header may not match expected format
Expected: ID,Name,Email,PreferredGame,SkillLevel,PreferredRole,PersonalityScore,PersonalityType
Found: ID,Name,Email,Phone,PreferredGame,SkillLevel,PreferredRole,PersonalityScore,PersonalityType
CSV Load Summary:
- Successfully loaded: 100 participants
- Errors/Skipped: 0 lines
- Total valid participants: 100
Successfully loaded 100 participants from CSV file
Saving teams to output CSV...
Saving teams to: formed_teams-.csv
Successfully saved 25 teams to: formed_teams-.csv
SUCCESS: Saved 25 teams to 'formed_teams-.csv'

Sample of loaded data (first 3 participants):
- P001 - Participant_1 (Chess) | Skill: 2 | Role: Strategist | Type: Leader
- P002 - Participant_2 (FIFA) | Skill: 5 | Role: Attacker | Type: Balanced
- P003 - Participant_3 (FIFA) | Skill: 4 | Role: Defender | Type: Thinker
```

## Concurrency-Demonstration

```
private static void demonstrateConcurrency(){ 1 usage 2 YenuIQ808
    System.out.println("\n7. BASIC CONCURRENCY DEMONSTRATION");
    System.out.println("-----");

    List<Participant> participants = Arrays.asList(
        new Participant( id: "P001", name: "Alice", email: "alice@edu.com", phoneNumber: "0771753109", preferredGame: "Valorant", skillLevel: 8, preferredRole: "Strategist", personalityScore: 95),
        new Participant( id: "P002", name: "Bob", email: "bob@edu.com", phoneNumber: "0791953109", preferredGame: "CS:GO", skillLevel: 6, preferredRole: "Attacker", personalityScore: 75),
        new Participant( id: "P003", name: "Charlie", email: "charlie@edu.com", phoneNumber: "0781853109", preferredGame: "DOTA 2", skillLevel: 7, preferredRole: "Defender", personalityScore: 65)
    );
    System.out.println("Starting team formation in background thread...");

    Thread backgroundThread = new Thread(() -> {
        try {
            Thread.sleep( millis: 1000);
            TeamBuilder builder = new TeamBuilder(participants, teamSize: 2);
            List<Team> teams = builder.formBalancedTeams();
            System.out.println("✓ Background thread completed: Formed " + teams.size() + " teams");
        } catch (Exception e) {
            System.out.println("✗ Background thread error: " + e.getMessage());
        }
    });

    backgroundThread.start();

    System.out.println("Main thread continues working while teams form in background...");
    System.out.println("Doing other work...");

    try {
        backgroundThread.join();
        System.out.println("✓ Concurrency demonstration completed!");
    } catch (InterruptedException e) {
        System.out.println("Thread was interrupted");
    }
}
```

### 7. BASIC CONCURRENCY DEMONSTRATION

-----

Starting team formation in background thread...

Main thread continues working while teams form in background...

Doing other work...

✓ Background thread completed: Formed 2 teams

✓ Concurrency demonstration completed!

## Complete-organizer-workflow

```
private static void demonstrateOrganizerWorkflow(){ 1 usage  Yenuli0808
    System.out.println("\n8. ORGANIZER WORKFLOW DEMONSTRATION");
    System.out.println("=====");

    try {
        Organizer organizer = new Organizer( id: "ORG001", name: "Tournament Manager");

        organizer.uploadCSV( filename: "participants_sample.csv");
        organizer.setFormationParameters(4);
        organizer.runTeamFormation();
        organizer.viewFormationResults();
        organizer.generateAdvancedReport();
        organizer.saveTeams( filename: "organized_teams-.csv");

        System.out.println("\n ORGANIZER WORKFLOW COMPLETED!");
    } catch (Exception e) {
        System.out.println("Organizer workflow failed: " + e.getMessage());
    }
}
```

```
8. ORGANIZER WORKFLOW DEMONSTRATION
=====
Organizer created: Tournament Manager (ID: ORG001)

ORGANIZER ACTION: Uploading CSV file...
Loading from: participants_sample.csv
Warning: CSV header may not match expected format
Expected: ID,Name,Email,PreferredGame,SkillLevel,PreferredRole,PersonalityScore,PersonalityType
Found: ID,Name,Email,Phone,PreferredGame,SkillLevel,PreferredRole,PersonalityScore,PersonalityType
CSV Load Summary:
- Successfully loaded: 100 participants
- Errors/Skipped: 0 lines
- Total valid participants: 100
SUCCESS: Uploaded 100 participants from participants_sample.csv

ORGANIZER ACTION: Setting formation parameters...
SUCCESS: Team size set to 4 players per team

ORGANIZER ACTION: Running team formation algorithm...
Applying Advanced Constraints:
1. One leader per team maximum
2. Max 2 players from same game
3. At least 3 different roles
4. Balanced personality mix
```

[illegible]

```
Dec 04, 2025 12:48:33 PM Service.CSVHandler loadParticipants
INFO: Loading participants from: participants_sample.csv (Source: LOCAL_FILE)
Dec 04, 2025 12:48:33 PM Service.CSVHandler loadParticipantsFromFile
INFO: Successfully loaded 100 participants from participants_sample.csv
Basic algorithm has constraint issues, trying advanced...
Applying Advanced Constraints:
1. One leader per team maximum
2. Max 2 players from same game
3. At least 3 different roles
4. Balanced personality mix
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
```

```
=== ROLE DIVERSITY CHECK ===
Team 1: 4 unique roles (ideal: 3) -  PASS
Team 2: 4 unique roles (ideal: 3) -  PASS
Team 3: 4 unique roles (ideal: 3) -  PASS
Team 4: 4 unique roles (ideal: 3) -  PASS
Team 5: 4 unique roles (ideal: 3) -  PASS
Team 6: 4 unique roles (ideal: 3) -  PASS
Team 7: 4 unique roles (ideal: 3) -  PASS
```

```
ORGANIZER ACTION: Viewing formation results...
=== TEAM FORMATION RESULTS ===

Team T1: Team 1 (4/4 members) | Avg Skill: 6.0
Games: [Chess, DOTA 2, CS:GO, FIFA]
Roles: [Strategist, Attacker, Coordinator, Supporter]
Personalities: 2 Leaders, 1 Balanced, 1 Thinkers

Team T2: Team 2 (4/4 members) | Avg Skill: 6.0
Games: [Basketball, FIFA, Valorant]
Roles: [Supporter, Strategist, Coordinator, Attacker]
Personalities: 2 Leaders, 1 Balanced, 1 Thinkers
```

ORGANIZER ACTION: Generating advanced team report...

=== ADVANCED TEAM ANALYSIS ===

Constraints Applied: Leader limit, Game variety, Role diversity, Personality mix

Team 1:

Players: 4/4

Avg Skill: 6.0

Personalities: 2L 1B 1T

Games: [Chess, DOTA 2, CS:GO, FIFA]

Roles: [Strategist, Attacker, Coordinator, Supporter]

Constraints: xLeader ✓Games ✓Roles xPersonality

Team 2:

Players: 4/4

Avg Skill: 6.0

Personalities: 2L 1B 1T

Games: [Basketball, FIFA, Valorant]

Roles: [Supporter, Strategist, Coordinator, Attacker]

Constraints: xLeader ✓Games ✓Roles xPersonality

ORGANIZER ACTION: Saving teams to CSV...

Saving teams to: organized\_teams-.csv

Successfully saved 25 teams to: organized\_teams-.csv

SUCCESS: Saved 25 teams to organized\_teams-.csv

ORGANIZER WORKFLOW COMPLETED!

# Unit-and-Integration-Test(JUnit)

## ConcurrentProcessingTest

```
@Test @Yenuli0808
public void testConcurrentTeamFormation() throws Exception{
    List<Participant> participants = createTestParticipants();
    TeamBuilder teamBuilder = new TeamBuilder(participants, teamSize: 2);

    System.out.println("starting concurrent team formation...");

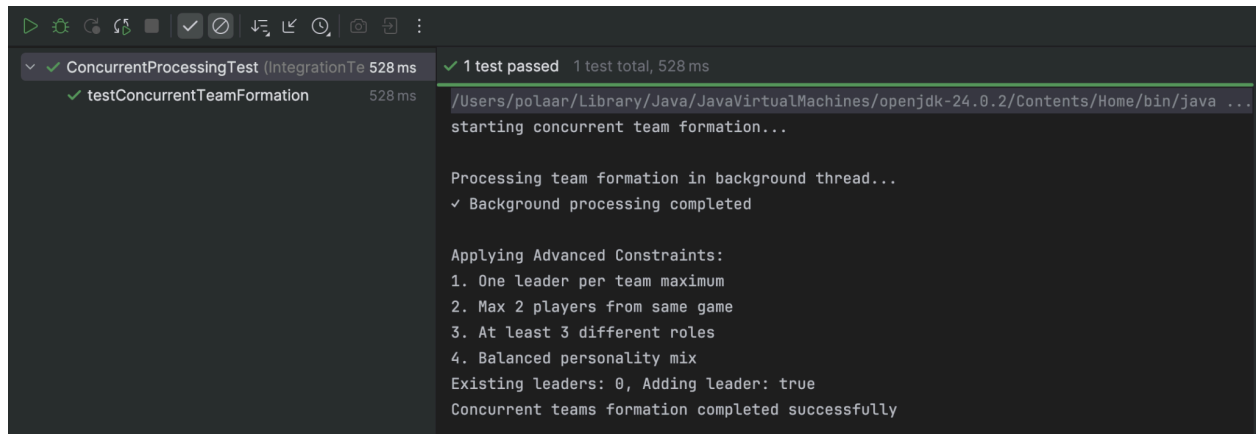
    //starting concurrent team formation
    CompletableFuture<List<Team>> future = teamBuilder.formTeamsConcurrently();
    assertFalse( message: "Future should not complete immediately", future.isDone());

    List<Team> teams = future.get( timeout: 5, TimeUnit.SECONDS); // THIS WILL WAIT FOR COMPLETION WITH TIMEOUT

    assertNotNull( message: "teams should not be null", teams);
    assertFalse( message: "Teams should not be empty", teams.isEmpty());
    assertEquals( message: "Should form 2 teams from 4 participants", expected: 2, teams.size());

    System.out.println("Concurrent teams formation completed successfully");

    teamBuilder.shutdown();
}
```



The screenshot shows an IDE interface with a test runner. The top bar indicates "1 test passed" and "1 test total, 528 ms". The test name "testConcurrentTeamFormation" is listed with a duration of "528 ms". The output pane on the right displays the following text:

```
/Users/polaar/Library/Java/JavaVirtualMachines/openjdk-24.0.2/Contents/Home/bin/java ...
starting concurrent team formation...

Processing team formation in background thread...
✓ Background processing completed

Applying Advanced Constraints:
1. One leader per team maximum
2. Max 2 players from same game
3. At least 3 different roles
4. Balanced personality mix
Existing leaders: 0, Adding leader: true
Concurrent teams formation completed successfully
```

## OrganizerParticipantIntegrationTest

```
21 public void testOrganizerCreatesTeams_ParticipantViewsThem() throws IOException {
22     //creating temporary CSV file for organizer to load
23     File tempFile = createTestCSVFile();
24
25     Organizer organizer = new Organizer( id: "ORG001", name: "Test Organizer");
26     List<Participant> participants = organizer.uploadCSV(tempFile.getAbsolutePath());
27
28     organizer.setFormationParameters(3);
29     List<Team> teams = organizer.runTeamFormation();
30
31     ParticipantPortal portal = new ParticipantPortal();
32     portal.initializeData(teams, participants);
33
34     // Test: Participants can view their teams
35     int participantsWithTeams = 0;
36     for (Participant participant : participants) {
37         try {
38             Team team = portal.viewTeamAssignment(participant.getId());
39             if (team != null) {
40                 participantsWithTeams++;
41                 assertTrue( message: "Team should contain participant",
42                     team.getMembers().contains(participant));
43             }
44         } catch (Exception e) {
45             // Some participants might not be assigned due to team size constraints
46             System.out.println("Participant " + participant.getId() + " not assigned: " + e.getMessage());
47         }
48     }
49     System.out.println("✓ Integration test passed: " + participantsWithTeams +
50         "/" + participants.size() + " participants can view their teams");
51
52     tempFile.delete();
53 }
```

```
✓ OrganizerParticipantIntegrationTest (Int 49 ms) ✓ 1 test passed 1 test total, 49 ms
  ✓ testOrganizerCreatesTeams_ParticipantViewsThem 49 ms
/Users/polaar/Library/Java/JavaVirtualMachines/openjdk-24.0.2/Contents/Home/bin/java ...
Created test CSV file: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/integration_test4441892343476306147.csv
Organizer created: Test Organizer (ID: ORG001)

ORGANIZER ACTION: Uploading CSV file...
Dec 04, 2025 3:33:08 PM Service.CSVHandler loadParticipants
INFO: Loading participants from: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/integration_test4441892343476306147.csv (Source: L0)
Loading from: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/integration_test4441892343476306147.csv
CSV Load Summary:
- Successfully loaded: 10 participants
- Errors/Skipped: 0 lines
- Total valid participants: 10
Dec 04, 2025 3:33:08 PM Service.CSVHandler loadParticipantsFromFile
INFO: Successfully loaded 10 participants from /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/integration_test4441892343476306147.c
SUCCESS: Uploaded 10 participants from /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/integration_test4441892343476306147.csv

ORGANIZER ACTION: Setting formation parameters...
SUCCESS: Team size set to 3 players per team
```

```
ORGANIZER ACTION: Running team formation algorithm...
Applying Advanced Constraints:
1. One leader per team maximum
2. Max 2 players from same game
3. At least 3 different roles
4. Balanced personality mix
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Basic algorithm has constraint issues, trying advanced...
Applying Advanced Constraints:
1. One leader per team maximum
2. Max 2 players from same game
3. At least 3 different roles
4. Balanced personality mix
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true
Existing leaders: 0, Adding leader: true

=== ROLE DIVERSITY CHECK ===
Team 1: 3 unique roles (ideal: 3) -  PASS
Team 2: 3 unique roles (ideal: 3) -  PASS
Team 3: 2 unique roles (ideal: 3) -  IDEAL NOT MET
Team 4: 2 unique roles (ideal: 3) -  IDEAL NOT MET
SUCCESS: Formed 4 balanced teams
Initializing with 4 teams and 10 participants
Created 10 participant-team mappings
✓ Integration test passed: 10/10 participants can view their teams
```

## PersonalityClassifierTest

```
16      @Test  @Yenuli0808
17      public void testPersonalityClassification_BoundaryValues(){
18          assertEquals(PersonalityType.LEADER, PersonalityClassifier.classify( score: 90));
19          assertEquals(PersonalityType.LEADER, PersonalityClassifier.classify( score: 100));
20          assertEquals(PersonalityType.BALANCED, PersonalityClassifier.classify( score: 70));
21          assertEquals(PersonalityType.BALANCED, PersonalityClassifier.classify( score: 89));
22          assertEquals(PersonalityType.THINKER, PersonalityClassifier.classify( score: 50));
23          assertEquals(PersonalityType.THINKER, PersonalityClassifier.classify( score: 69));
24      }
25
26      @Test  @Yenuli0808
27      public void testPersonalityClassification_InvalidScores(){
28          assertThrows(IllegalArgumentException.class, () -> PersonalityClassifier.classify( score: 45));
29          assertThrows(IllegalArgumentException.class, () -> PersonalityClassifier.classify( score: 101));
30      }
31
32      @Test  @Yenuli0808
33      public void testSurveyScoreCalculation(){
34          int[] responses = {5,4,5,4,5};
35          int score = PersonalityClassifier.calculateFromSurvey(responses);
36
37          assertEquals( expected: 92, score);
38          assertEquals(PersonalityType.LEADER, PersonalityClassifier.classify(score));
39      }
```

```
40
41      @Test  @Yenuli0808
42      public void testInvalidSurveyResponses(){
43          int[] invalidResponses = {6,4,5,4,5};
44
45          Exception exception = assertThrows(IllegalArgumentException.class, () -> {
46              PersonalityClassifier.calculateFromSurvey(invalidResponses);
47          });
48          assertTrue(exception.getMessage().contains("Responses must be 1-5"));
49      }
50
```

## TeamBuilderTest

```
7      @Test  @Yenuli0808
8      public void testTeamFormation_Basic(){
9          List<Participant> participants = createTestParticipants();
10         TeamBuilder builder = new TeamBuilder(participants, teamSize: 3);
11
12         List<Team> teams = builder.formBalancedTeams();
13
14         assertNotNull(teams);
15         assertFalse(teams.isEmpty());
16         assertEquals( expected: 2, teams.size());
17     }
```

```

8
9  @Test & YenulIO808
10 public void testTeamFormation_AdvancedConstraints(){
11     List<Participant> participants = createTestParticipants();
12     TeamBuilder builder = new TeamBuilder(participants, teamSize: 3);
13
14     List<Team> teams = builder.formAdvancedTeams();
15     assertNotNull(teams);
16
17     for(Team team : teams){
18         assertTrue( condition: team.countPersonalityType(PersonalityType.LEADER)<=1);
19         assertTrue( condition: team.getUniqueGames().size() >=1);
20         assertTrue( condition: team.getUniqueRoles().size() >= Math.min(3, team.getUniqueGames().size()));
21     }
22 }

```

```

@Test & YenulIO808 *
public void testPersonalityScoring_FirstLeaderGetsHighScore(){
    List<Participant> participants = Arrays.asList(
        //Leaders
        new Participant( id: "P001", name: "Leader1", email: "Leader1@edu.com", phoneNumber: "0711111111", preferredGame: "Valorant", skillLevel: 8, preferredRole: "Strategist", personalityScore: 95),
        new Participant( id: "P002", name: "Leader2", email: "Leader2@edu.com", phoneNumber: "0722222222", preferredGame: "CS:GO", skillLevel: 7, preferredRole: "Attacker", personalityScore: 92),

        //Thinkers
        new Participant( id: "P003", name: "Thinker1", email: "thinker1@edu.com", phoneNumber: "0733333333", preferredGame: "DOTA 2", skillLevel: 6, preferredRole: "Defender", personalityScore: 65),
        new Participant( id: "P004", name: "Thinker2", email: "thinker2@edu.com", phoneNumber: "0744444444", preferredGame: "FIFA", skillLevel: 5, preferredRole: "Supporter", personalityScore: 60),

        //Balanced
        new Participant( id: "P005", name: "Balanced1", email: "balanced1@edu.com", phoneNumber: "0755555555", preferredGame: "Basketball", skillLevel: 7, preferredRole: "Coordinator", personalityScore: 80),
        new Participant( id: "P006", name: "Balanced2", email: "balanced2@edu.com", phoneNumber: "0766666666", preferredGame: "Valorant", skillLevel: 6, preferredRole: "Attacker", personalityScore: 78)
    );

    TeamBuilder builder = new TeamBuilder(participants, teamSize: 3);
    List<Team> teams = builder.formAdvancedTeams();

    assertEquals( message: "Should form 2 teams from 6 participants", expected: 2, teams.size());

    //checking each team composition
    for(Team team : teams){
        int leaders = team.countPersonalityType(PersonalityType.LEADER);
        int thinkers = team.countPersonalityType(PersonalityType.THINKER);
        int balanced = team.countPersonalityType(PersonalityType.BALANCED);

        assertTrue( message: "Each team should have 0-1 leaders", condition: leaders <= 1);
        assertTrue( message: "Each team should have 1-2 thinkers", condition: thinkers >= 1 && thinkers <= 2);
        assertTrue( message: "Each team should have balanced players", condition: balanced >= 1);

        assertTrue( message: "Each team should have multiple roles", condition: team.getUniqueRoles().size() >= 2);
    }
}

```

```

@Test & YenulIO808
public void testRoleScoring_NewRoleGetsHighScore() {
    List<Participant> participants = Arrays.asList(
        new Participant( id: "P001", name: "P1", email: "p1@edu.com", phoneNumber: "0711111111", preferredGame: "Valorant", skillLevel: 8, preferredRole: "Strategist", personalityScore: 85),
        new Participant( id: "P002", name: "P2", email: "p2@edu.com", phoneNumber: "0722222222", preferredGame: "CS:GO", skillLevel: 7, preferredRole: "Attacker", personalityScore: 80),
        new Participant( id: "P003", name: "P3", email: "p3@edu.com", phoneNumber: "0733333333", preferredGame: "DOTA 2", skillLevel: 6, preferredRole: "Defender", personalityScore: 75),
        new Participant( id: "P004", name: "P4", email: "p4@edu.com", phoneNumber: "0744444444", preferredGame: "FIFA", skillLevel: 5, preferredRole: "Supporter", personalityScore: 70),
        new Participant( id: "P005", name: "P5", email: "p5@edu.com", phoneNumber: "0755555555", preferredGame: "Basketball", skillLevel: 6, preferredRole: "Coordinator", personalityScore: 78)
    );

    TeamBuilder builder = new TeamBuilder(participants, teamSize: 5);
    List<Team> teams = builder.formAdvancedTeams();

    assertNotNull(teams);
    assertFalse(teams.isEmpty());

    // Assert that all teams have good role diversity
    for (Team formedTeam : teams) {
        int roleCount = formedTeam.getUniqueRoles().size();
        assertTrue( message: "Each team should have at least 3 roles, but got " + roleCount,
            condition: roleCount >= 3);
    }
}

```

## CSVHandlerTest

```
@Test
public void testCSVParsing() throws IOException {
    File tempFile = File.createTempFile( prefix: "test_participants", suffix: ".csv");
    tempFile.deleteOnExit();

    try(FileWriter writer = new FileWriter(tempFile)) {
        writer.write( str: "ID,Name,Email,PreferredGame,SkillLevel,PreferredRole,PersonalityScore,PersonalityType\n");
        writer.write( str: "P101,Participant_101,user101@university.edu,Valorant,8,Strategist,85,Balanced\n");
        writer.write( str: "P102,Participant_102,user102@university.edu,CS:GO,6,Attacker,75,Balanced\n");
    }

    CSVHandler csvHandler = new CSVHandler();
    List<Participant> participants = csvHandler.loadParticipants(tempFile.getAbsolutePath());

    assertEquals( expected: 2, participants.size());
    assertEquals( expected: "P101", participants.get(0).getId());
    assertEquals( expected: "Valorant", participants.get(0).getPreferredGame());
}
```

```
✓ CSVHandlerTest (ServiceTests) 39 ms ✓ 1 test passed 1 test total, 39 ms
✓ testCSVParsing 39 ms
/Users/polaar/Library/Java/JavaVirtualMachines/openjdk-24.0.2/Contents/Home/bin/java ...
Dec 04, 2025 3:21:17 PM Service.CSVHandler loadParticipants
INFO: Loading participants from: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/test_participants3707359196021462958.csv
Loading from: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/test_participants3707359196021462958.csv
CSV Load Summary:
- Successfully loaded: 2 participants
- Errors/Skipped: 0 lines
- Total valid participants: 2
Dec 04, 2025 3:21:17 PM Service.CSVHandler loadParticipantsFromFile
INFO: Successfully loaded 2 participants from /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/test_participants37073591960
```

```
@Test
public void testTeamExport() throws IOException {
    List<Participant> participants = Arrays.asList(
        new Participant( id: "P001", name: "Alice", email: "alice@edu.com", phoneNumber: "0711111111",
            preferredGame: "Valorant", skillLevel: 8, preferredRole: "Strategist", personalityScore: 95),
        new Participant( id: "P002", name: "Bob", email: "bob@edu.com", phoneNumber: "0722222222",
            preferredGame: "CS:GO", skillLevel: 6, preferredRole: "Attacker", personalityScore: 75)
    );

    TeamBuilder builder = new TeamBuilder(participants, teamSize: 2);
    List<Team> teams = builder.formBalancedTeams();

    File tempFile = File.createTempFile( prefix: "test_teams", suffix: ".csv");
    tempFile.deleteOnExit();

    CSVHandler csvHandler = new CSVHandler();
    csvHandler.saveTeamsToCSV(teams, tempFile.getAbsolutePath());

    assertTrue(tempFile.exists());
    assertTrue( condition: tempFile.length() > 0);
}
```

```
✓ CSVHandlerTest (ServiceTests) 45 ms ✓ 1 test passed 1 test total, 45 ms
✓ testTeamExport 45 ms
/Users/polaar/Library/Java/JavaVirtualMachines/openjdk-24.0.2/Contents/Home/bin/java ...
Dec 04, 2025 3:22:26 PM Service.CSVHandler saveTeamsToCSV
Saving teams to: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/test_teams14345517086076526895.csv
INFO: Saving 1 teams to: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/test_teams14345517086076526895.csv
Successfully saved 1 teams to: /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/test_teams14345517086076526895.csv
Dec 04, 2025 3:22:26 PM Service.CSVHandler saveTeamsToFile
INFO: Successfully saved 1 teams to /var/folders/22/bmq9v6yj2nz2cnw7kjqg5gk00000gn/T/test_teams14345517086076526895.csv
```

```
@Test  @Yenuli0808
public void testCSVHandler_InvalidFile() {
    CSVHandler handler = new CSVHandler();
    assertThrows(FileNotFoundException.class, () -> {
        handler.loadParticipants( filename: "nonexistent_file.csv");
    });
}
```

|                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ✓ CSVHandlerTest (ServiceTests) 25 ms |       | ✓ 1 test passed 1 test total, 25 ms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ✓ testCSVHandler_InvalidFile          | 25 ms | <pre>/Users/polaar/Library/Java/JavaVirtualMachines/openjdk-24.0.2/Contents/Home/bin/java ... Dec 04, 2025 3:22:54 PM Service.CSVHandler loadParticipants INFO: Loading participants from: nonexistent_file.csv (Source: LOCAL_FILE) Loading from: nonexistent_file.csv Dec 04, 2025 3:22:54 PM Service.CSVHandler loadParticipantsFromFile SEVERE: CSV file not found: nonexistent_file.csv Please ensure the file exists in one of these locations: - nonexistent_file.csv - data/nonexistent_file.csv - Project root directory</pre> |

# 7.Appendix

## Commits

|                                                                                                                     |                                                                                               |                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  main                              |  All users |  All time                                                                                    |
| Commits on Dec 4, 2025                                                                                              |                                                                                               |                                                                                                                                                                                 |
| final touchup in codes (commenting)                                                                                 |                                                                                               |                                                                                                                                                                                 |
|  Yenuli0808 committed 1 hour ago   |                                                                                               |                                                                                                                                                                                 |
|                                                                                                                     |                                                                                               | d9bc2dc   |
| sequence no commenting updated                                                                                      |                                                                                               |                                                                                                                                                                                 |
|  Yenuli0808 committed 16 hours ago |                                                                                               |                                                                                                                                                                                 |
|                                                                                                                     |                                                                                               | c2cee33   |
| Commits on Dec 3, 2025                                                                                              |                                                                                               |                                                                                                                                                                                 |
| sequence diagram commenting for complete CSV                                                                        |                                                                                               |                                                                                                                                                                                 |
|  Yenuli0808 committed yesterday    |                                                                                               |                                                                                                                                                                                 |
|                                                                                                                     |                                                                                               | e7be6fb   |
| Commits on Dec 2, 2025                                                                                              |                                                                                               |                                                                                                                                                                                 |
| sequence comment updating                                                                                           |                                                                                               |                                                                                                                                                                                 |
|  Yenuli0808 committed 2 days ago   |                                                                                               |                                                                                                                                                                                 |
|                                                                                                                     |                                                                                               | 834ce25   |
| Commits on Nov 29, 2025                                                                                             |                                                                                               |                                                                                                                                                                                 |
| sequence no commenting updates for run team formation                                                               |                                                                                               |                                                                                                                                                                                 |
|  Yenuli0808 committed 5 days ago   |                                                                                               |                                                                                                                                                                                 |
|                                                                                                                     |                                                                                               | 3b2b79e   |
| Commits on Nov 26, 2025                                                                                             |                                                                                               |                                                                                                                                                                                 |
| run team formation sequence diagram number commenting                                                               |                                                                                               |                                                                                                                                                                                 |
|  Yenuli0808 committed last week    |                                                                                               |                                                                                                                                                                                 |
|                                                                                                                     |                                                                                               | 220aa65   |
| sequence number commenting for upload CSV sequence                                                                  |                                                                                               |                                                                                                                                                                                 |
|  Yenuli0808 committed last week    |                                                                                               |                                                                                                                                                                                 |
|                                                                                                                     |                                                                                               | 282b5b0   |

|                                                                                                                      |  |                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commits on Nov 26, 2025                                                                                              |  |                                                                                                                                                                                     |
| Create test class with 4 unit testing                                                                                |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | d4b4079   |
| add new test case testFixedAlgorithm_MeetsAllRequirements()                                                          |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 423a205   |
| Update TeamBuilderTest.java                                                                                          |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 3a9ecac   |
| updated test classes                                                                                                 |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 2e61a43   |
| Update TeamBuilderTest.java                                                                                          |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 1bbf496   |
| updated classes after modification on skill balance                                                                  |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 385cdbb   |
| Commits on Nov 24, 2025                                                                                              |  |                                                                                                                                                                                     |
| Update Organizer.java                                                                                                |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 7375082   |
| updated classes                                                                                                      |  |                                                                                                                                                                                     |
|  Yenuli0808 committed last week   |  |                                                                                                                                                                                     |
|                                                                                                                      |  | e126cf2   |
| Commits on Nov 23, 2025                                                                                              |  |                                                                                                                                                                                     |
| Update TeamTest.java                                                                                                 |  |                                                                                                                                                                                     |
|  Yenuli0808 committed 2 weeks ago |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 6535b77   |
| Update CSVHandlerTest.java                                                                                           |  |                                                                                                                                                                                     |
|  Yenuli0808 committed 2 weeks ago |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 4618641   |
| Create ConcurrentProcessingTest.java                                                                                 |  |                                                                                                                                                                                     |
|  Yenuli0808 committed 2 weeks ago |  |                                                                                                                                                                                     |
|                                                                                                                      |  | b30e2ac   |
| Create TeamConstraintsTest.java                                                                                      |  |                                                                                                                                                                                     |
|  Yenuli0808 committed 2 weeks ago |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 566afd9   |
| Update DefaultTeamConstraints.java                                                                                   |  |                                                                                                                                                                                     |
|  Yenuli0808 committed 2 weeks ago |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 76831fe   |
| Update DefaultTeamConstraints.java                                                                                   |  |                                                                                                                                                                                     |
|  Yenuli0808 committed 2 weeks ago |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 1772c86   |
| Update Main.java                                                                                                     |  |                                                                                                                                                                                     |
|  Yenuli0808 committed 2 weeks ago |  |                                                                                                                                                                                     |
|                                                                                                                      |  | 4fa2876   |

|                                       |                                  |                                                                                                                                                                                 |
|---------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commits on Nov 22, 2025               |                                  |                                                                                                                                                                                 |
| Update TestDataFactory.java           | Yenuli0808 committed 2 weeks ago | a242b40   |
| Update OrganizerWorkflowTest.java     | Yenuli0808 committed 2 weeks ago | 3c697e1   |
| Create TestDataFactory.java           | Yenuli0808 committed 2 weeks ago | 130efc3   |
| Create OrganizerWorkflowTest.java     | Yenuli0808 committed 2 weeks ago | 3b4cc09   |
| Create CSVHandlerTest.java            | Yenuli0808 committed 2 weeks ago | 351e31e   |
| Create TeamBuilderTest.java           | Yenuli0808 committed 2 weeks ago | 486fe50   |
| Create PersonalityClassifierTest.java | Yenuli0808 committed 2 weeks ago | 4658899   |
| Create ParticipantTest.java           | Yenuli0808 committed 2 weeks ago | 78785cb   |
| Create TeamTest.java                  | Yenuli0808 committed 2 weeks ago | 286ec28   |
| Update TeamBuilder.java               | Yenuli0808 committed 2 weeks ago | cf332eb   |
| updated main and teamBuilder classes  | Yenuli0808 committed 2 weeks ago | ea7e863   |
| updated classes                       | Yenuli0808 committed 2 weeks ago | 8672a36   |
| updated main and organizer classes    | Yenuli0808 committed 2 weeks ago | 98c02d9   |

|                                      |                                  |                                                                                                                                                                                     |
|--------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commits                              |                                  |                                                                                                                                                                                     |
| main                                 |                                  |                                                                                                                                                                                     |
| All users                            |                                  |                                                                                                                                                                                     |
| All time                             |                                  |                                                                                                                                                                                     |
| Commits on Nov 22, 2025              |                                  |                                                                                                                                                                                     |
| updated main and CSV handler classes | Yenuli0808 committed 2 weeks ago | 7dcfe23   |
| Update CSVHandler.java               | Yenuli0808 committed 2 weeks ago | 9a3e5f0   |
| Update Constant.java                 | Yenuli0808 committed 2 weeks ago | f131276   |
| Update CSVHandler.java               | Yenuli0808 committed 2 weeks ago | 49f1ae6   |
| Update Constant.java                 | Yenuli0808 committed 2 weeks ago | 17ca02d   |
| Update participants_sample.csv       | Yenuli0808 committed 2 weeks ago | 25b4461   |
| Commits on Nov 21, 2025              |                                  |                                                                                                                                                                                     |
| Update Survey.java                   | Yenuli0808 committed 2 weeks ago | 5baa63b   |
| Update BaseEntity.java               | Yenuli0808 committed 2 weeks ago | 2cdc556   |
| Updated files                        | Yenuli0808 committed 2 weeks ago | 69d6349   |
| updated 2 classes and CSV file       | Yenuli0808 committed 2 weeks ago | 1abd97b   |
| Updated main and sample.csv          | Yenuli0808 committed 2 weeks ago | f550b35   |
| update 2 files                       | Yenuli0808 committed 2 weeks ago | f1b50fd   |

|                                                                    |         |                                                                                                                                                                         |
|--------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commits on Nov 20, 2025                                            |         |                                                                                                                                                                         |
| update 3 files<br>Yenuli0808 committed 2 weeks ago                 | b767064 |   |
| Update Organizer.java<br>Yenuli0808 committed 2 weeks ago          | 92a7b34 |   |
| update 3 files<br>Yenuli0808 committed 2 weeks ago                 | 08b172e |   |
| Update Main.java<br>Yenuli0808 committed 2 weeks ago               | 9802b74 |   |
| Update participants_sample.csv<br>Yenuli0808 committed 2 weeks ago | e2d1000 |   |
| Commits on Nov 19, 2025                                            |         |                                                                                                                                                                         |
| Update Main.java<br>Yenuli0808 committed 2 weeks ago               | e69393c |   |
| Commits on Nov 18, 2025                                            |         |                                                                                                                                                                         |
| Update Main.java<br>Yenuli0808 committed 2 weeks ago               | 9741411 |   |
| Update TeamBuilder.java<br>Yenuli0808 committed 2 weeks ago        | 11123c2 |   |
| Update TeamBuilder.java<br>Yenuli0808 committed 2 weeks ago        | f803799 |   |

|                                                            |         |                                                                                                                                                                             |
|------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Commits on Nov 16, 2025                                    |         |                                                                                                                                                                             |
| update these 4<br>Yenuli0808 committed 3 weeks ago         | b0bcb7a |       |
| update 2 classes<br>Yenuli0808 committed 3 weeks ago       | dbf5fd9 |   |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago       | dfc53bd |   |
| update these 2 classes<br>Yenuli0808 committed 3 weeks ago | 3abaf9f |   |
| add these csv files<br>Yenuli0808 committed 3 weeks ago    | 36ce35d |   |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago       | 2dcb688 |   |
| Update CSVHandler.java<br>Yenuli0808 committed 3 weeks ago | a23bd3d |   |
| Update Organizer.java<br>Yenuli0808 committed 3 weeks ago  | e08a79d |   |
| update 2 classes<br>Yenuli0808 committed 3 weeks ago       | be55b1f |   |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago       | 0d94424 |   |
| Update CSVHandler.java<br>Yenuli0808 committed 3 weeks ago | 4015172 |   |
| Update Organizer.java<br>Yenuli0808 committed 3 weeks ago  | acd126e |   |
| Commits on Nov 15, 2025                                    |         |                                                                                                                                                                             |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago       | 66c78e9 |   |
| Update Organizer.java<br>Yenuli0808 committed 3 weeks ago  | 78d3faa |   |

Commits

|                                                             |           |          |
|-------------------------------------------------------------|-----------|----------|
| main                                                        | All users | All time |
| Commits on Nov 15, 2025                                     |           |          |
| update these 5 files<br>Yenuli0808 committed 3 weeks ago    | 1f28aff   |          |
| Update Survey.java<br>Yenuli0808 committed 3 weeks ago      | cddc1d6   |          |
| update Participant.java<br>Yenuli0808 committed 3 weeks ago | a988e41   |          |
| Commits on Nov 14, 2025                                     |           |          |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago        | 929ec2e   |          |
| Commits on Nov 12, 2025                                     |           |          |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago        | 8d97eb3   |          |

|                                                                    |         |  |
|--------------------------------------------------------------------|---------|--|
| Commits on Nov 11, 2025                                            |         |  |
| Create formed_teams.csv<br>Yenuli0808 committed 3 weeks ago        | 0052acb |  |
| Create organized_teams.csv<br>Yenuli0808 committed 3 weeks ago     | 1da5a0a |  |
| Create participants_sample.csv<br>Yenuli0808 committed 3 weeks ago | 2a058e2 |  |
| Update Organizer.java<br>Yenuli0808 committed 3 weeks ago          | fd255fc |  |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago               | be67c06 |  |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago               | c1c9a7e |  |
| Commits on Nov 10, 2025                                            |         |  |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago               | a1bb3f8 |  |
| Update Main.java<br>Yenuli0808 committed 3 weeks ago               | 680305c |  |
| Update Survey.java<br>Yenuli0808 committed 3 weeks ago             | a81286d |  |
| Create Organizer.java<br>Yenuli0808 committed 3 weeks ago          | f39247d |  |
| Update Survey.java<br>Yenuli0808 committed 3 weeks ago             | 5af1731 |  |
| Create 2425432.iml<br>Yenuli0808 committed 3 weeks ago             | da8db91 |  |
| Update CSVHandler.java<br>Yenuli0808 committed 3 weeks ago         | b2d9228 |  |
| Commits on Nov 9, 2025                                             |         |  |
| Update Main.java<br>Yenuli0808 committed last month                | 595920d |  |
| Update Participant.java<br>Yenuli0808 committed last month         | 88e7d07 |  |
| Update FormatTable.java<br>Yenuli0808 committed last month         | 3c7d27d |  |
| Update Team.java<br>Yenuli0808 committed last month                | 79c2483 |  |

|                                     |                                 |                                                                                                |
|-------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------|
| Commits on Nov 8, 2025              |                                 |                                                                                                |
| Update TeamBuilder.java             | Yenuli0808 committed last month | 201f1ed  <> |
| Create TeamFormationConstraint.java | Yenuli0808 committed last month | 04bde48  <> |
| Create DefaultTeamConstraints.java  | Yenuli0808 committed last month | ace858d  <> |
| Create Survey.java                  | Yenuli0808 committed last month | 1a1f01b  <> |
| Update Participant.java             | Yenuli0808 committed last month | 3e66f85  <> |
| Update FormatTable.java             | Yenuli0808 committed last month | e245add  <> |
| Update PersonalityClassifier.java   | Yenuli0808 committed last month | 2e84075  <> |
| Commits on Nov 6, 2025              |                                 |                                                                                                |
| Update Main.java                    | Yenuli0808 committed last month | e9ca61c  <> |
| Update TeamBuilder.java             | Yenuli0808 committed last month | af84c33  <> |
| Commits on Nov 5, 2025              |                                 |                                                                                                |
| Create CSVHandler.java              | Yenuli0808 committed last month | c47ca92  <> |
| Update TeamBuilder.java             | Yenuli0808 committed last month | 977e4b0  <> |
| Update Main.java                    | Yenuli0808 committed last month | e3f56b9  <> |
| Update FormatTable.java             | Yenuli0808 committed last month | 0fb4c59  <> |

|                                             |                                 |                                                                                                  |
|---------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------|
| Commits on Nov 5, 2025                      |                                 |                                                                                                  |
| Create TeamFormationStrategy.java           | Yenuli0808 committed last month | 9ee1317  <> |
| Update Participant.java                     | Yenuli0808 committed last month | d839085  <> |
| Update Team.java                            | Yenuli0808 committed last month | 1bf3e60  <> |
| Update Participant.java                     | Yenuli0808 committed last month | 565350c  <> |
| Create FormatTable.java                     | Yenuli0808 committed last month | 61223c1  <> |
| Create BaseEntity.java                      | Yenuli0808 committed last month | e59c630  <> |
| Update Main.java                            | Yenuli0808 committed last month | db2e482  <> |
| Update Team.java                            | Yenuli0808 committed last month | 0ceb6b6  <> |
| Update Team.java                            | Yenuli0808 committed last month | 7992d3d  <> |
| Create Constant.java                        | Yenuli0808 committed last month | 06a1b04  <> |
| Update TeamBuilder.java                     | Yenuli0808 committed last month | 9a30321  <> |
| Update Team.java                            | Yenuli0808 committed last month | 4240a84  <> |
| Create TeamBuilder.java                     | Yenuli0808 committed last month | c37ee02  <> |
| Create PersonalityClassifier.java           | Yenuli0808 committed last month | c0a95c5  <> |
| Update Participant.java                     | Yenuli0808 committed last month | b5e30f7  <> |
| Create PersonalityType.java                 | Yenuli0808 committed last month | 16354ca  <> |
| Update Team.java                            | Yenuli0808 committed last month | f730610  <> |
| Add Team class to Gaming_Club_Model package | Yenuli0808 committed last month | 5b5d4e4  <> |

|                                                                                                                                                                                                                                 |                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Create .gitignore</b><br> Yenuli0808 committed last month                                                                                   | bc30c99   |
| <b>Create .gitignore</b> <br> Yenuli0808 committed last month | 3a6e8ee   |
| <b>Create Participant.java</b><br> Yenuli0808 committed on Nov 4                                                                               | 0c5f5d0   |
| <b>Initial commit with IntelliJ project setup</b><br> Yenuli0808 committed on Nov 4                                                            | 2c77743   |